CHAPTER 7

Femur and Pelvic Girdle

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CONTENTS

RADIOGRAPHIC ANATOMY

Lower Limb (Extremity)

In Chapter 6, three groups of bones of the lower limb—the foot, lower leg, and distal femur—were described, along with the associated knee and ankle joints (Fig. 7.1).

The lower limb bones discussed in this chapter are the **proximal femur** and the **pelvic girdle.** The joints involving these two groups of bones, also included in this chapter, are the important **hip joint** and the **sacroiliac** and **symphysis pubis** joints of the pelvic girdle.

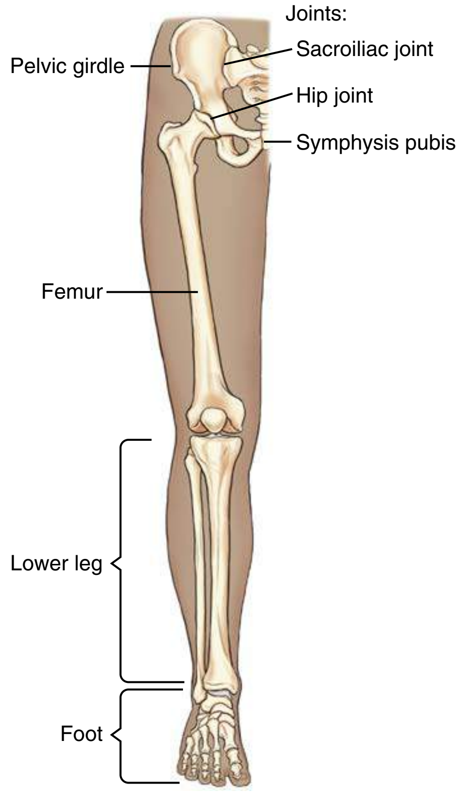

Fig. 7.1 Lower limb.

FEMUR

The **femur** is the longest and strongest bone in the body. The entire weight of the body is transferred through this bone and the associated joints at each end. Therefore, these joints are a frequent source of pathology when trauma occurs. The anatomy of the mid- to distal femur was discussed in Chapter 6.

Proximal Femur

The proximal femur consists of four essential parts, the head (1), neck (2), and greater (3) and lesser trochanters (4) *(tro-kan′-ters).*

The **head** of the femur is rounded and smooth for articulation with the hip bones. It contains a depression, or pit, near its center called the ***fovea capitis*** *(fo′-ve-ah cap′-i-tis),* wherein a major ligament called the ***ligament of the head of the femur,*** or the ***ligament capitis femoris,*** is attached to the head of the femur.

The **neck** of the femur is a strong pyramidal process of bone that connects the head with the body or shaft in the region of the trochanters.

The **greater trochanter** is a large prominence located **superiorly** and **laterally** to the femoral shaft and is palpable as a bony landmark. The **lesser trochanter** is a smaller, blunt, conical eminence that projects **medially** and **posteriorly** from the junction of the neck and shaft of the femur. The trochanters are joined posteriorly by a thick ridge called the **intertrochanteric** *(in″-ter-tro″-kan-ter′-ik)* **crest.** The **body** or **shaft** of the femur is long and almost cylindrical (Fig. 7.2).

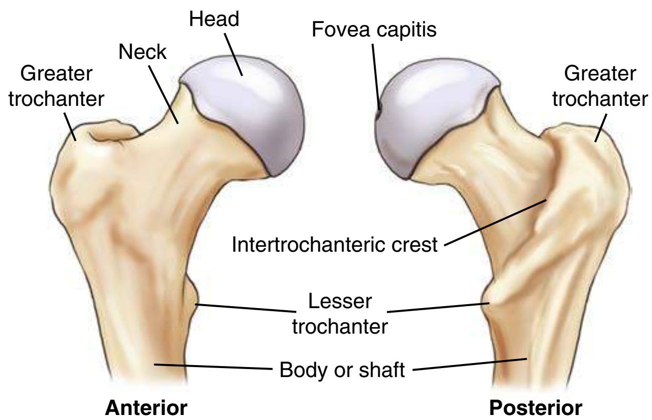

Fig. 7.2 Proximal femur.

Angles of the Proximal Femur The angle of the neck to the shaft on an average adult is approximately **125°,** with a variance of ±15°, depending on the width of the pelvis and the length of the lower limbs. For example, in a long-legged person with a narrow pelvis, the femur would be nearer vertical, which then would change the angle of the neck to about 140°. This angle would be less (110° to 115°) for a shorter person with a wider pelvis.

On an average adult in the anatomic position, the longitudinal plane of the femur is approximately **10° from vertical,** as shown on the left in Fig. 7.3. This vertical angle is nearer 15° on someone with a wide pelvis and shorter limbs and only about 5° on a long-legged person. This angle affects positioning and the central ray (CR) angles for a lateral knee, as described in Figs. 6.19 and 6.20.

Another angle of the neck and head of the femur that is important in radiography is the **15° to 20° anterior angle** of the head and neck in relation to the body of the femur (see right drawing of Fig. 7.3). The head projects somewhat anteriorly or forward as a result of this angle. This angle becomes important in radiographic positioning; the femur and lower leg must be rotated **15° to 20° internally** to place the femoral neck parallel to the image receptor (IR) for a true anteroposterior (AP) projection of the proximal femur.

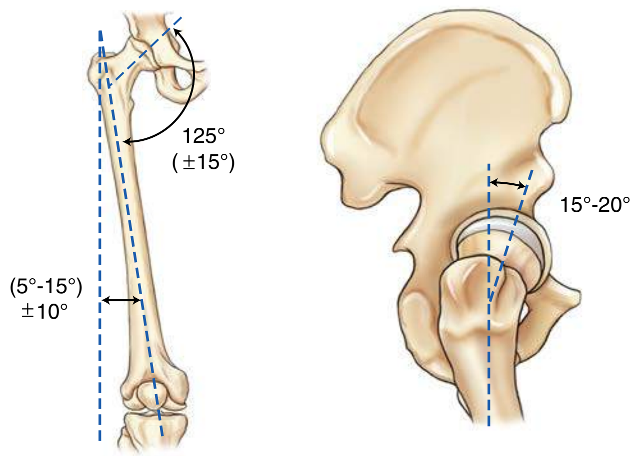

Fig. 7.3 Angles of proximal femur.

PELVIS

The complete **pelvis** (meaning a *basin*) serves as the base of the trunk and forms the connection between the vertebral column and lower limbs. The pelvis consists of four bones—two **hip bones** (also called **innominate bones**), one **sacrum** *(sa′-krum),* and one **coccyx** *(kok′-siks)* (Fig. 7.4). The sacrum articulates superiorly with the fifth lumbar vertebra to form the lumbosacral joint (also called the L5–S1 joint). The right and left hip (iliac) bones articulate posteriorly with the sacrum to form the sacroiliac joints.[1]

NOTE: The sacrum and the coccyx also are considered parts of the distal vertebral column and in this textbook are discussed in Chapter 9, along with the lumbar spine.

HIP BONE

Each hip bone is composed of three divisions: (1) **ilium** *(il′-e-um),* (2) **ischium** *(is′-ke-um),* and (3) **pubis** *(pu′-bis).* In a child, these three divisions are separate bones, but they fuse into one bone during the middle teens. The fusion occurs in the area of the **acetabulum** *(as″-e-tab′-u-lum).* The acetabulum is a deep, cup-shaped cavity that accepts the head of the femur to form the hip joint (Fig. 7.5).

The ilium, the largest of the three divisions, is located superior to the acetabulum. The ischium is inferior and posterior to the acetabulum, whereas the pubis is inferior and anterior to the acetabulum. Each of these three parts is described in detail in the following sections.

Ilium

Each **ilium** is composed of a **body** and an **ala,** or wing (Fig. 7.6). The body of the ilium is the more inferior portion near the acetabulum and includes the superior two-fifths of the acetabulum. The ala, or wing portion, is the thin and flared superior part of the ilium.

The **crest** of the ilium is the superior margin of the ala; it extends from the **anterior superior iliac spine** (ASIS) to the **posterior superior iliac spine** (PSIS). In radiographic positioning, the uppermost peak of the crest often is referred to as the **iliac crest,** but it actually extends between the ASIS and the PSIS.

Below the ASIS is a less prominent projection referred to as the **anterior inferior iliac spine.** Similarly, inferior to the PSIS is the **posterior inferior iliac spine.**

Positioning Landmarks The two important positioning landmarks of these borders and projections are the **iliac crest** and the **ASIS.**

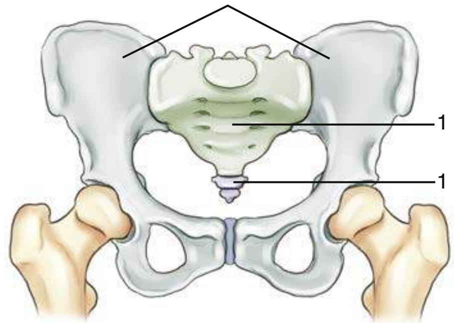

Fig. 7.4 Pelvis—four bones: two hip bones, sacrum, and coccyx.

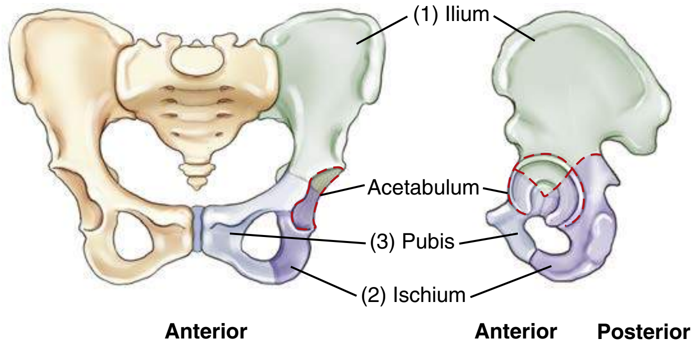

Fig. 7.5 Hip bone—three parts.

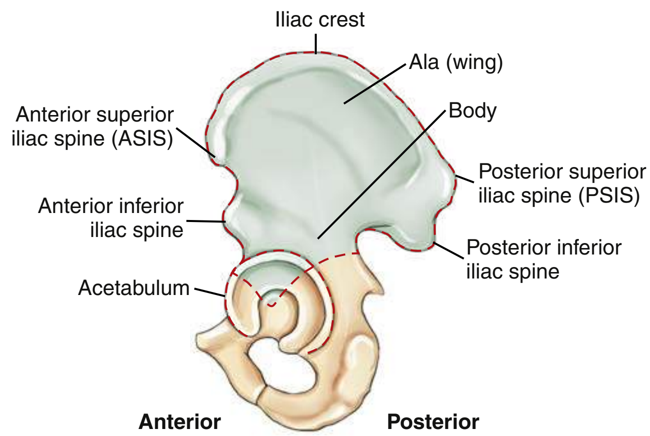

Fig. 7.6 Ilium.

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Ischium

The **ischium** is that part of the hip bone that lies inferior and posterior to the acetabulum. Each ischium is divided into a **body** and a **ramus** (Fig. 7.7). The superior portion of the body of the ischium makes up the posteroinferior two-fifths of the acetabulum. The lower portion of the body of the ischium (formerly called the *superior ramus*) projects caudally and medially from the acetabulum, ending at the **ischial tuberosity.** Projecting anteriorly from the ischial tuberosity is the **ramus** of the ischium.

The rounded roughened area near the junction of the lower body and the inferior rami is a landmark called the **tuberosity** of the ischium, or the **ischial** *(is′-ke-al)* **tuberosity.**

Posterior to the acetabulum is a bony projection termed the ***ischial spine***. A small part of the ischial spine also is visible on a frontal view of the pelvis, as shown in Fig. 7.8. (It is also seen in the anatomy review radiograph; see Fig. 7.16.)

Directly superior to the ischial spine is a deep notch termed the ***greater sciatic notch.*** Inferior to the ischial spine is a smaller notch termed the ***lesser sciatic notch.***

Positioning Landmarks The ischial tuberosities bear most of the weight of the body when an individual sits. They can be palpated through the soft tissues of each buttock in a prone position. However, because of discomfort and possible embarrassment to the patient, this landmark is not used as commonly as the previously described ASIS and crest of the ilium.

Pubis

The last of the three divisions of one hip bone is the **pubis,** or **pubic bone.** The **body** of the pubis is anterior and inferior to the acetabulum and includes the anteroinferior one-fifth of the acetabulum.

Extending anteriorly and medially from the body of each pubis is a **superior ramus.** The two superior rami meet in the midline to form an amphiarthrodial joint, the **symphysis pubis** *(sim′-fi-sis pu′-bis),* which also is correctly called the **pubic symphysis.** Each **inferior ramus** passes down and posterior from the symphysis pubis to join the ramus of the respective ischium.

The **obturator foramen** *(ob′-tu-ra″-tor fo-ra′-men)* is a large opening formed by the ramus and body of each ischium and by the pubis. The obturator foramen is the largest foramen in the human skeletal system.

Positioning Landmark The crests of the ilium and ASIS are important positioning landmarks. The superior margin of the symphysis pubis is a possible landmark for pelvis and hip positioning and for positioning of the abdomen, because it defines the inferior margin of the abdomen. However, if other associated landmarks are available, the symphysis pubis generally is not used as a palpated landmark because of patient modesty and potential embarrassment.

SUMMARY OF TOPOGRAPHIC LANDMARKS

Important positioning landmarks of the pelvis are reviewed in Fig. 7.9. The most superior aspects of the **iliac crest** and the **ASIS** are easily palpated. The ASIS is one of the more frequently used positioning landmarks of the pelvis. It also is commonly used to check for rotation of the pelvis and/or lower abdomen by determination of whether the distance between the ASIS and the tabletop is equal on both sides.

The **greater trochanter** of the femur can be located by firm palpation of the soft tissues of the upper thigh. Note that the prominence of the greater trochanter is at about the same level as the superior border of the **symphysis pubis,** whereas the **ischial tuberosity** is 1½ to 2 inches (4 to 5 cm) below the symphysis pubis. These distances vary between a male and a female pelvis because of general differences in shape, as described later in this chapter.

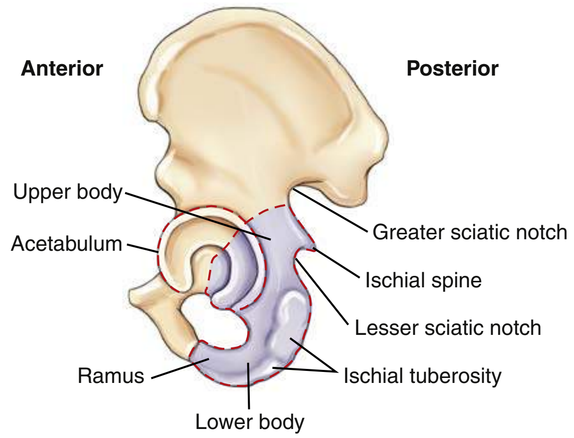

Fig. 7.7 Ischium.

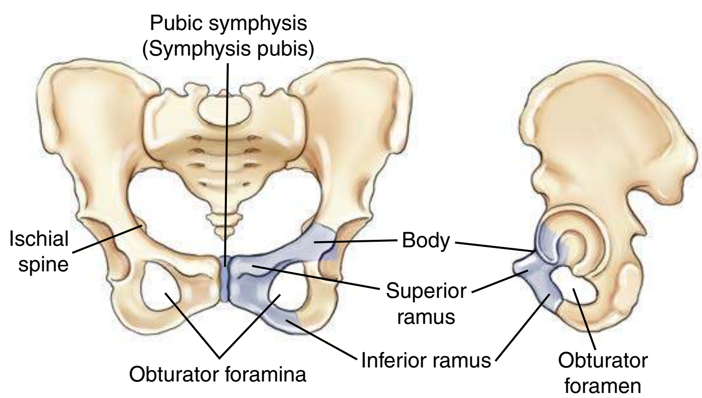

Fig. 7.8 Pubis (pubic bone).

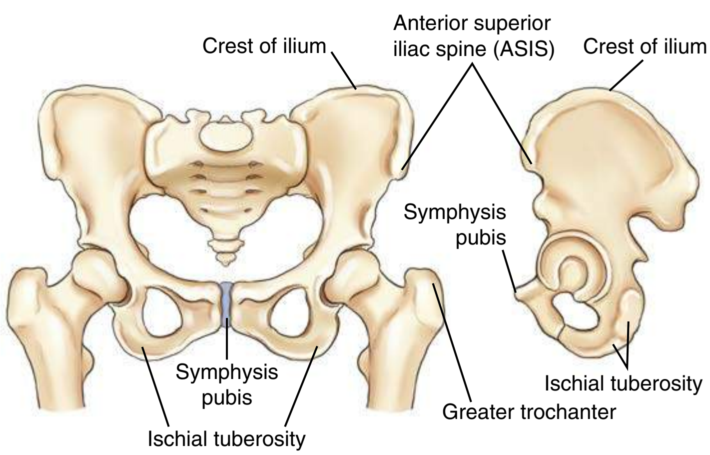

Fig. 7.9 Bony topographic landmarks of the pelvis.

TRUE AND FALSE PELVIS

A plane through the **brim** of the pelvis divides the pelvic area into two cavities. The pelvic brim is defined by the superior portion of the symphysis pubis anteriorly and by the superior, prominent part of the sacrum posteriorly. The general area above or superior to the oblique plane through the pelvic brim is termed the ***greater,*** or ***false, pelvis.*** The flared portion of the pelvis, which is formed primarily by the alae, or wings, of the ilia, forms the lateral and posterior limits of the false pelvis, whereas the abdominal muscles of the anterior wall define the anterior limits. The lower abdominal organs rest on the floor of the greater pelvis, as does the fetus within a pregnant uterus.

The area inferior to a plane through the pelvic brim is termed the ***lesser,*** or ***true, pelvis.*** The true pelvis is a cavity that is completely surrounded by bony structures. The size and shape of the true pelvis are of greatest importance during the birth process because the **true pelvis forms the actual birth canal** (Fig. 7.10).

True Pelvis

The oblique plane defined by the brim of the pelvis is termed the ***inlet,*** or ***superior aperture,*** of the true pelvis. The **outlet,** or **inferior aperture,** of the true pelvis is defined by the two ischial tuberosities and the tip of the coccyx (Fig. 7.11). The three sides of the triangularly shaped outlet are formed by a line between the ischial tuberosities and a line between each ischial tuberosity and the coccyx. The area between the inlet and outlet of the true pelvis is termed the ***cavity*** of the true pelvis. During the birth process, the baby must travel through the inlet, cavity, and outlet of the true pelvis.

Birth Canal

During a routine delivery, the baby's head first travels through the pelvic inlet, then to the midcavity, and finally through the outlet before it exits in a forward direction, as shown in Fig. 7.12.

Because of sensitivity of the fetus to radiation, radiographs of the pelvis generally are **not** taken during pregnancy. If the dimensions of the birth canal of the pelvis are in question, certain ultrasound procedures can be done to evaluate for potential problems during the birth process.

Pelvic Ring The pelvic ring is a term applied in orthopedics describing the sturdy, ringlike structure formed by the union of the ilium, ischium, and pubic bones, along with the sacrum and coccyx. This term is often used in described specific fractures that can disrupt the alignment of these bones.[2]

Male Versus Female Pelvis Differences[3]

There are four common shapes of the human pelvic inlet:

- Gynecoid: round
- Platypelloid: wider right to left than anterior to posterior
- Android: heart shaped
- Anthropoid: wider anterior to posterior than right to left

The general shape of the female pelvis is different enough from that of the male pelvis to enable discrimination of one from the other on pelvis radiographic images. In general, the **female pelvis** is wider, with the ilia more flared and more shallow from front to back (gynecoid or platypelloid). The **male pelvis** is narrower, deeper (anthropoid), or less flared with a heart-shaped pelvic inlet (android). In overall appearance on a frontal view, the female pelvis is wider with a round pelvic inlet. Therefore, the first difference between the male pelvis and female pelvis is the difference in the **overall general shape** of the entire pelvis, along with the shape differences of the pelvic inlet (Fig. 7.13).

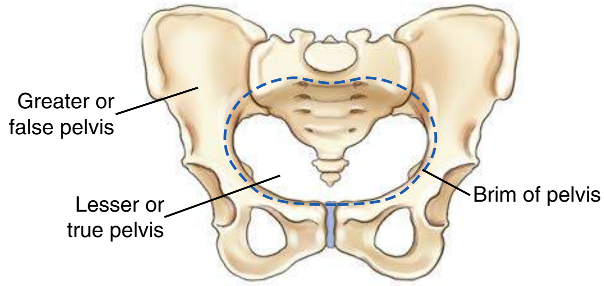

Fig. 7.10 Pelvic cavities.

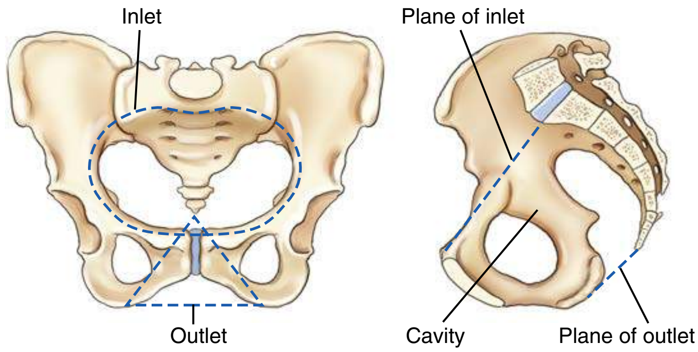

Fig. 7.11 Lesser or true pelvis.

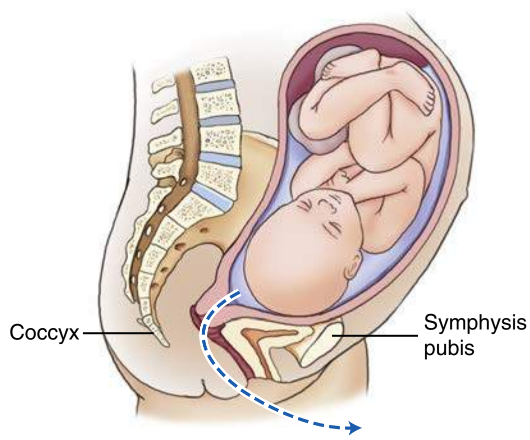

Fig. 7.12 Birth canal—sagittal sectional view.

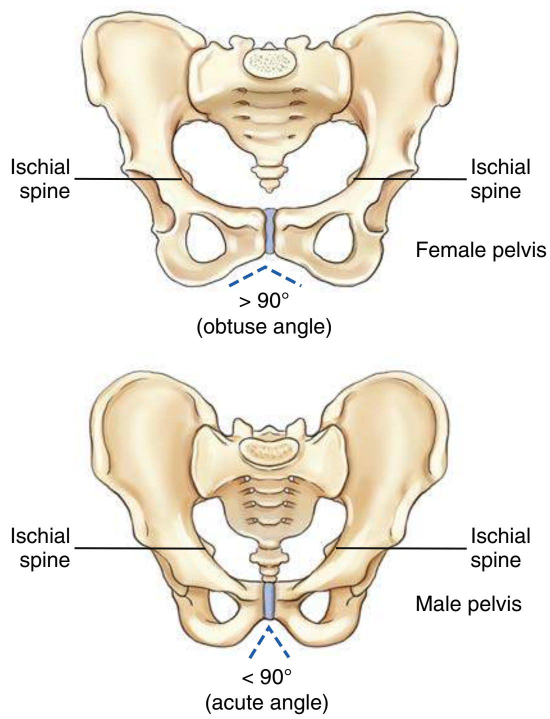

Fig. 7.13 Pelvis—male versus female.

A second major difference is the **angle of the pubic arch,** formed by the inferior rami of the pubis just inferior to the symphysis pubis. In the female, this angle is between 80° and 85°, whereas in the male, the pubic arch usually forms an acute angle between 50° and 60°.

A third difference is that the ischial spines generally do not project as far medially toward the pelvic cavity in the female as they do in males. They are more visible along the lateral margins of the pelvic cavity in the male than in the female in the AP pelvis projection.

NOTE: The general shape of the pelvis does vary considerably from one individual to another, so the pelvis of a slender female may resemble a male pelvis. In general, however, the differences are usually obvious enough that the gender of the patient can be determined from a radiographic image of the pelvis.

See Table 7.1 for a summary of male and female pelvic characteristics.

TABLE 7.1 SUMMARY OF MALE AND FEMALE PELVIC CHARACTERISTICS

	MALE	FEMALE
General shape (shape of pelvic inlet)	Narrower, deeper, less flared. Pelvic inlet is more oval or heart shaped (anthropoid or android)	Wider, more shallow, more flared. Pelvic inlet rounder (gynecoid or platypelloid)
Angle of pubic arch	Narrower angle (50° to 60°)	Wider angle (80° to 85°)
Ischial spines	More protrusion into pelvic inlet (or cavity)	Less protrusion into pelvic inlet

Fig. 7.14 Female pelvis.

Fig. 7.15 Male pelvis.

Male Versus Female Pelvis Radiographs

Figs. 7.14 and 7.15 are pelvic radiographs of a female subject and a male subject, respectively. Note the three differences between the typical female pelvis and typical male pelvis.

1. In overall shape, the male pelvis appears narrower and deeper and has a less-flared appearance of the ilia. The shape of the inlet on the male pelvis is not as large or as rounded as that of the female pelvis.
2. The pubic arch of the male pelvis has a smaller angle compared to the greater angle on the female pelvis. This angle is commonly one of the more noticeable differences.
3. Radiographic presence of the ischial spines along the lateral margins of the pelvic cavity is less pronounced with the female pelvis.

REVIEW EXERCISE WITH RADIOGRAPHS

Key pelvic anatomy is labeled on the AP pelvis radiograph of Fig. 7.16. A good review exercise is to cover up the answers (listed below) while identifying the labeled parts.

A. Iliac crest
B. ASIS (anterior end of crest)
C. Body of left ischium
D. Ischial tuberosity
E. Symphysis pubis (pubic symphysis)
F. Inferior ramus of right pubis
G. Superior ramus of right pubis
H. Right ischial spine
I. Acetabulum of right hip
J. Neck of right femur
K. Greater trochanter of right femur
L. Head of right femur
M. Ala, or wing, of right ilium

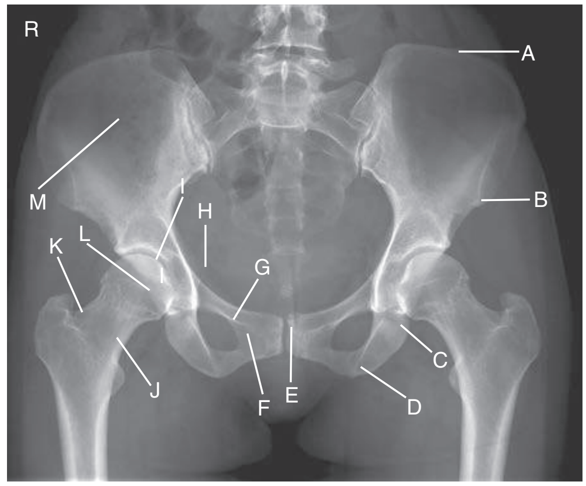

Fig. 7.16 AP Pelvis.

Lateral Hip

Fig. 7.17 presents a lateral radiograph of the proximal femur and hip, taken with an axiolateral projection (Danelius-Miller method), as demonstrated by the positioning in Fig. 7.18. Answers to the labeled parts are as follows:

A. Acetabulum
B. Femoral head
C. Femoral neck
D. Shaft or body
E. Area of lesser trochanter
F. Area of greater trochanter
G. Ischial tuberosity

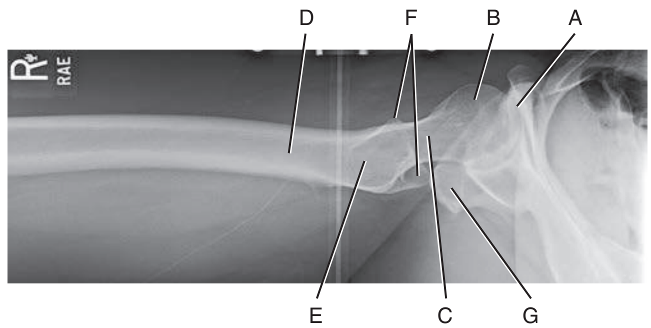

Fig. 7.17 Axiolateral projection.

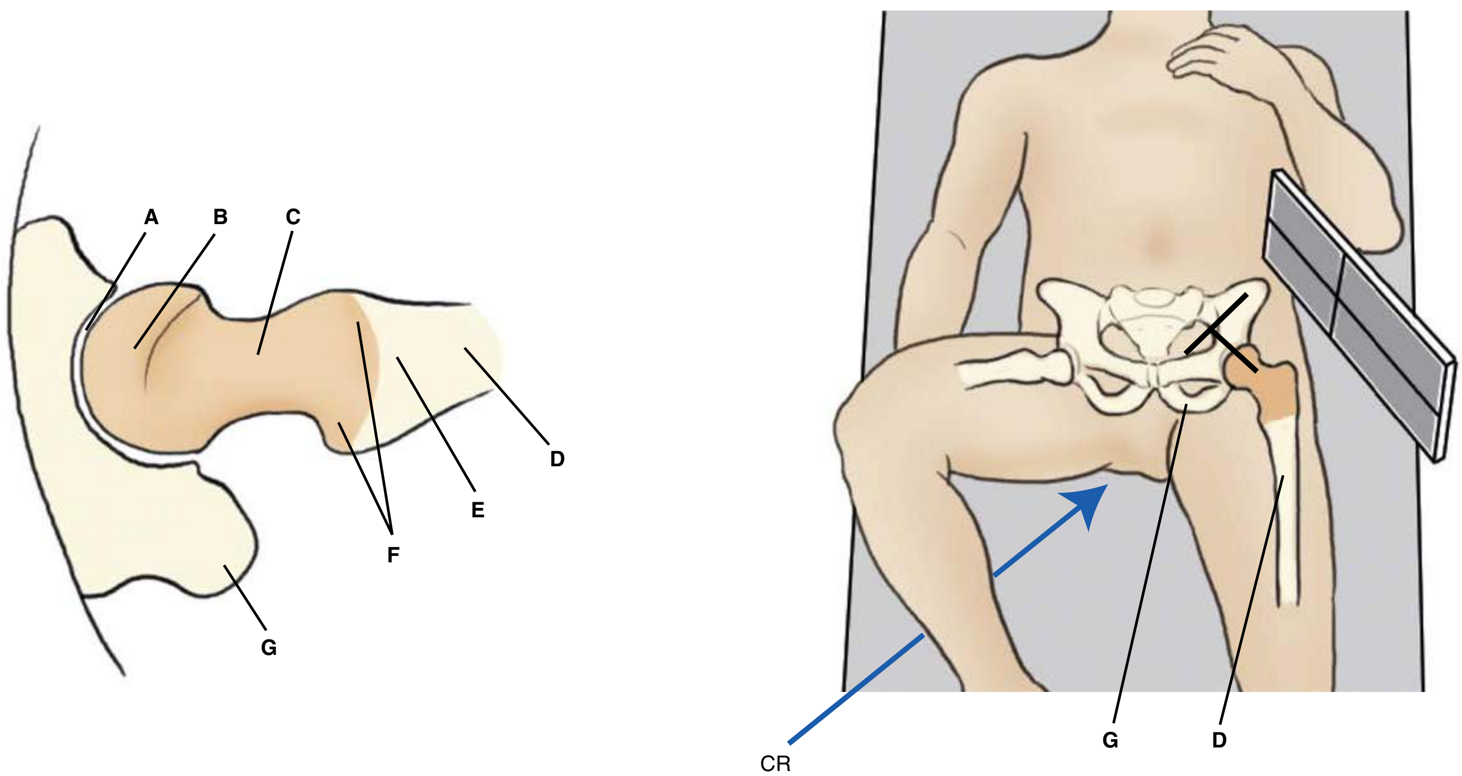

Fig. 7.18 Proximal femur and hip—lateral (axiolateral projection).

CLASSIFICATION OF JOINTS (FIG. 7.19)

The number of joints or articulations of the proximal femora and pelvis is limited, with the hip joint being the most obvious. These joints of the pelvis, in the following list, again are described according to their **classification, mobility type,** and **movement type.**

Sacroiliac joints: joints between the sacrum and each ilium
Symphysis pubis: structure between the right and left pubic bones
Union of acetabulum: temporary growth joint of each acetabulum that solidifies in the midteen years
Hip joints: joints between the head of the femur and the acetabulum of the pelvis

Sacroiliac Joints (Fig. 7.20)

The sacroiliac joints are wide flat joints located on each side obliquely between the sacrum and each ilium. These joints are situated at an unusual oblique angle, requiring special positioning to visualize the joint spaces radiographically.

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The sacroiliac joint is classified as a **synovial joint** in that it is enclosed in a fibrous articular capsule that contains synovial fluid. The bones are joined by firm sacroiliac ligaments. Generally, synovial joints by their nature are considered freely movable, or diarthrodial, joints. However, the sacroiliac joint is a special type of synovial joint that permits little movement. Due to its unique structure and movement, it's type of movement is **irregular gliding**. The reason for this classification is that the joint surfaces are very irregularly shaped and the interconnecting bones are snugly fitted because they serve a weight-bearing function. This shape restricts movement, and the cavity of the joint or the joint space may be reduced in size or even nonexistent in older persons, especially in males. Positioning of the sacroiliac joints is described in Chapter 9.

Symphysis Pubis

The symphysis pubis is the articulation of the right and left pubic bones located in the midline of the anterior pelvis. The most superior anterior aspect of this joint is palpable and is an important positioning landmark, as described earlier.

The symphysis pubis is classified as a **cartilaginous joint** of the **symphysis subtype** in that only limited movement is possible **(amphiarthrodial).** The two articular surfaces are separated by a fibrocartilaginous disk and are held together by certain ligaments. This interpubic disk of fibrocartilage is a relatively thick pad (thicker in females than males) that is capable of being compressed or partially displaced, thereby allowing limited movement of these bones, as in the case of pelvic trauma or during the childbirth process in females.

Union of Acetabulum

The three divisions of each hip bone are separate bones in a child but come together in the acetabulum by fusing during the middle teens to become completely indistinguishable in an adult. Therefore, this structure is classified as a **cartilaginous-type** joint of the **synchondrosis subtype,** which is **immovable,** or **synarthrodial,** in an adult. This joint is considered a temporary type of growth joint that is similar to the joints between the epiphyses and diaphyses of long bones in growing children.

Hip Joint

The hip joint is classified as a **synovial type,** which is characterized by a large fibrous capsule that contains synovial fluid. It is a **freely movable,** or **diarthrodial,** joint and is the truest example of a **ball and socket** (spheroidal) movement type. See Table 7.2 for a summary of the pelvic joints.

The head of the femur forms more than half a sphere as it fits into the relatively deep, cup-shaped acetabulum. This connection makes the hip joint inherently strong as it supports the weight of the body while still permitting a high degree of mobility. The articular capsule surrounding this joint is strong and dense, with the thickest part being superior, as would be expected because it is in line with the weight-bearing function of the hip joints. A series of strong bands of ligaments surround the articular capsule and joint in general, making this joint very strong and stable.

Movements of the hip joint include **flexion** and **extension, abduction** and **adduction, medial** (internal) and **lateral** (external) **rotation,** and **circumduction.**

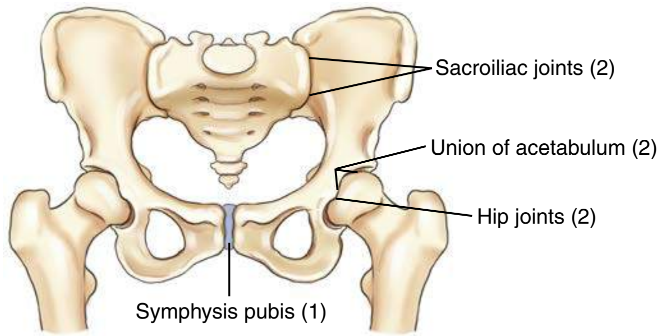


Fig. 7.19 Joints of pelvis.

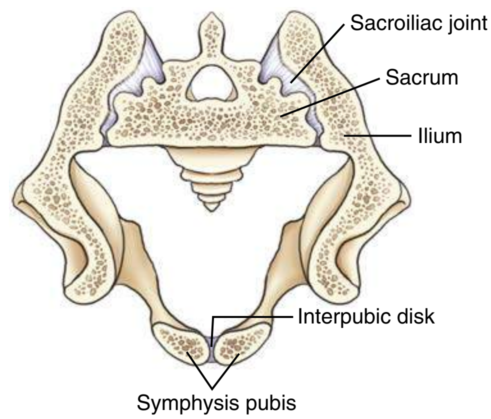


Fig. 7.20 Coronal view of a transverse section showing sacroiliac and symphysis pubis joints.

TABLE 7.2 SUMMARY OF PELVIC JOINTS

JOINTS	CLASSIFICATION	MOBILITY TYPE	MOVEMENT TYPE
Sacroiliac joint	Synovial	Limited movement	Irregular gliding
Symphysis pubis	Cartilaginous	Amphiarthrodial	Limited
Union of acetabulum	Cartilaginous	Synarthrodial (for adults)	Nonmovable
Hip joint	Synovial	Diarthrodial	Ball and socket (spheroidal)

RADIOGRAPHIC POSITIONING

Positioning Considerations

LOCATING THE FEMORAL HEAD AND NECK

A long-standing traditional method used to locate the femoral head and neck is first to determine the midpoint of a line between the ASIS and the symphysis pubis. The **neck** is approximately 2½ inches (6 to 7 cm), and the **head** 1½ inches (4 cm) distal and at right angles to the midpoint of this line (Figs. 7.21 and 7.22).

The **greater trochanters** are shown to be located on the same horizontal line as the symphysis pubis. However, the greater trochanters are difficult to palpate accurately on large or bariatric patients, and palpation of the symphysis pubis can be embarrassing for the patient. Therefore, a second method is suggested for locating the femoral head or neck that uses only the **ASIS,** which is easily palpated on all types of patients. The level of the symphysis pubis is between 3 and 4 inches (8 to 10 cm) inferior to the level of the ASIS. Therefore, the femoral neck can be readily located as being **1 to 2 inches (3 to 5 cm) medial and 3 to 4 inches (8 to 10 cm) distal to the ASIS.** This level also places it on the same horizontal plane as the symphysis pubis and the greater trochanters.

As was previously demonstrated, significant differences exist between the male pelvis and the female pelvis, but with some practice and allowances for male and female differences, both of these methods work well for locating the femoral head or neck for hip positioning.

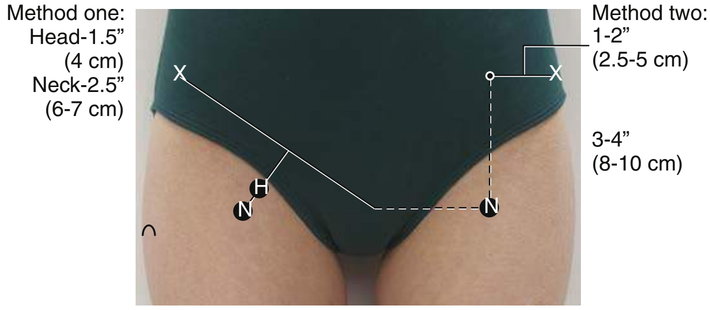


Fig. 7.21 Femoral head *(H)* or neck *(N)* localization.

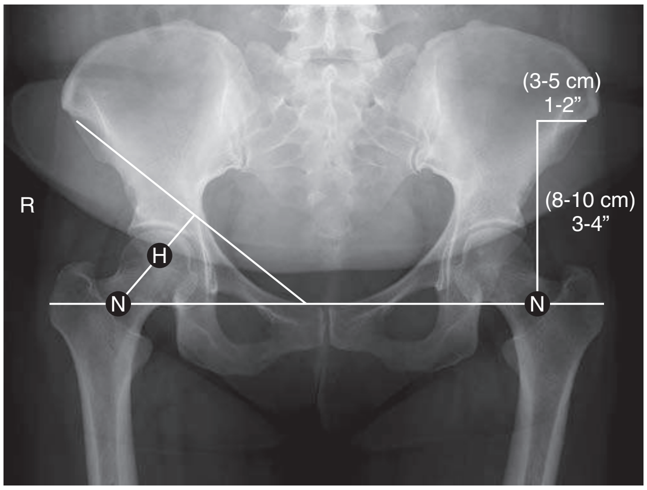


Fig. 7.22 Female pelvis, head *(H),* and neck *(N)* locations.

APPEARANCE OF PROXIMAL FEMUR IN ANATOMIC POSITION

As was described earlier in this chapter under anatomy of the proximal femur, the head and neck of the femur project approximately 15° to 20° anteriorly or forward with respect to the rest of the femur and the lower leg. Thus, when the leg is in the true anatomic position, as for a true AP leg, the proximal femur actually is rotated posteriorly 15° to 20° (Fig. 7.23). Therefore, the femoral neck appears shortened and the **lesser trochanter is visible** when the leg and ankle are truly AP, as in a true anatomic position.

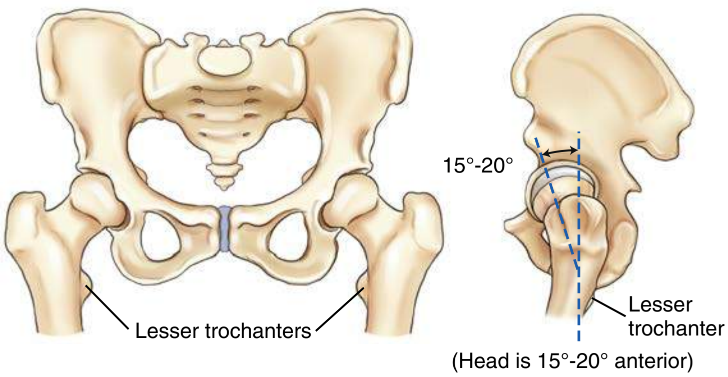


Fig. 7.23 Anatomic position (true AP of knee, leg, and ankle—but not of hip).

INTERNAL ROTATION OF LEG

By **internally rotating the entire lower limb,** the proximal femur and hip joint are positioned in a **true AP** projection. The neck of the femur is now parallel to the imaging surface and will not appear foreshortened.

The **lesser trochanter** is key in determining the correct leg and foot position (on a radiographic image). If the entire leg is rotated internally a full 15° to 20° (Fig. 7.24A), the outline of the lesser trochanter generally is not visible at all or is only slightly visible on some patients, when it is obscured by the shaft of the femur. If the leg is straight AP, or when it is externally rotated, the lesser trochanter is visible (see page 274).

EVIDENCE OF HIP FRACTURE

The femoral neck is a common fracture site for an older patient who has fallen. The typical physical sign for such a fracture is the **external rotation** of the involved foot, where the lesser trochanter is clearly visualized in profile, as can be seen on the left hip depicted in Fig. 7.32 and the drawing on the right (Fig. 7.24B). This radiographic sign is demonstrated on the following page (see Figs. 7.31 and 7.32).

WARNING: If evidence of a hip fracture is present (external foot rotation), a pelvis radiograph should be taken "as is" **without** attempting to rotate the leg internally, as would be necessary for a true AP hip projection.

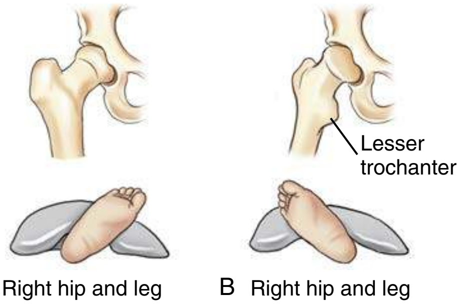


Fig. 7.24 A, Internal rotation (true AP of hip). B, External rotation (typical hip fracture position).

SUMMARY: EFFECT OF LOWER LIMB ROTATION

The photographs and associated pelvis radiographs on this page demonstrate the effects of lower limb rotation on the appearance of the proximal femora.

1. **Anatomic position** (Figs. 7.25 and 7.26)
   - Long axes of feet vertical
   - Femoral necks partially foreshortened
   - Lesser trochanters **partially visible**
2. **15° to 20° medial rotation** (desired position to visualize pelvis and hips; Figs. 7.27 and 7.28)
   - Long axes of feet and lower limbs rotated internally 15° to 20°
   - Femoral heads and necks in profile
   - True AP projection of proximal femora
   - Lesser trochanters **not visible** or only slightly visible on some patients
3. **External rotation** (Figs. 7.29 and 7.30)
   - Long axes of feet and lower limbs equally rotated laterally in a normal relaxed position
   - Femoral necks greatly foreshortened
   - Lesser trochanters visible in profile internally
4. Typical rotation with hip fracture (Figs. 7.31 and 7.32)
   - Long axis of left foot externally rotated (on side of hip fracture)
   - Unaffected right foot and limb in neutral position
   - Lesser trochanter on externally rotated (left) limb more visible; neck area foreshortened

Fig. 7.25 Anatomic position.

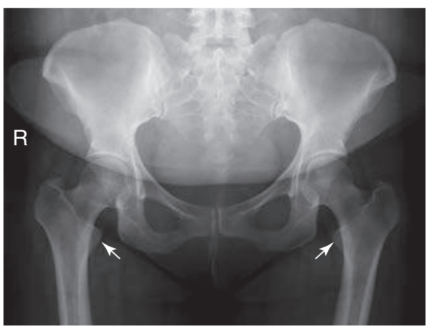


Fig. 7.26 1. Anatomic position without correction of lower limbs.

Fig. 7.27 15° to 20° medial rotation.

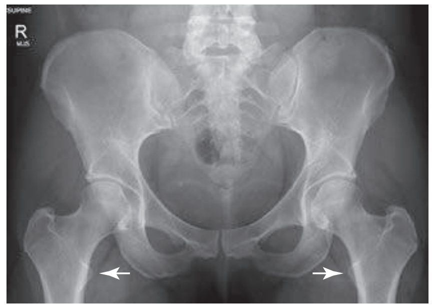


Fig. 7.28 2. 15° to 20° medial rotation.

Fig. 7.29 External rotation.

Fig. 7.30 3. External rotation.

Fig. 7.31 Typical rotation with hip fracture.

Fig. 7.32 4. Typical rotation with hip fracture.

SHIELDING GUIDELINES

Accurate gonadal shielding for pelvis and hip examinations is especially critical because of the proximity of radiation-sensitive gonads to the primary x-ray beam.

Male Shielding

Shielding is easier for males in that small contact shields, such as those shown in Fig. 7.33, can be used on **all males.** These shields are placed over the area of the testes without covering the essential anatomy of the pelvis or hips. However, care must be taken with pelvic radiographs that the top of the shield is placed at the **inferior margin of the symphysis pubis** to cover the testes adequately without obscuring the pubic and ischial areas of the pelvis.

Female Shielding

Ovarian contact shields for females require more critical placement to shield the area of the ovaries without covering essential pelvic or hip anatomy. Vinyl-covered lead material cut into various shapes and sizes can be used for this purpose for an AP pelvis or bilateral hip radiograph, as shown in Fig. 7.34. For a unilateral hip or proximal femur, larger contact shields can be used to cover the general pelvic area without covering the specific hip that is being examined, as shown in Fig. 7.35. Accurate location of the femoral head and neck makes this type of gonadal shielding possible.

Gonadal shielding may not be possible for females on certain AP pelvic projections in which the entire pelvis, including the sacrum and coccyx, must be demonstrated. Also, gonadal shielding may not be possible on lateral inferosuperior hip projections for males and females because shielding may obscure essential anatomy. However, **gonadal shielding should be used whenever possible for both males and females,** along with **close collimation** for all hip and pelvic projections. General pelvic trauma requiring visualization of the entire pelvis may prohibit ovarian shielding for females.

Exposure Factors and Patient Dose

To reduce total radiation dose to the patient, a higher kVp range of 80 to 90 may be used for hip and pelvic examinations. This higher kVp technique, with lower mAs, results in a lower radiation dose to the patient. Higher kVp, however, decreases subject contrast and may not be advisable, especially for older patients, who may have some loss of bone mass or density caused by osteoporosis; thus, they may require even lower kVp than average. Overexposure with high kVp on osteoporotic patients will decrease the visibility of the bony detail when using both analog and digital imaging systems.

Special Patient Considerations

PEDIATRIC APPLICATIONS

Pelvic and hip radiographic examinations are not performed often on children, except on newborns with developmental dysplasia of the hip (DDH). Correct shielding is especially important for infants and children because of the repeat radiographic examinations that are frequently required during the growth of the child. If holding the legs of an infant is required, an individual other than radiology personnel should do this while wearing a lead apron and lead gloves.

The degree and type of immobilization required for older children are dependent on the ability and willingness of the child to cooperate during the procedure. A mummy wrap (see Chapter 16) helps prevent the upper limbs from interfering with the anatomy of interest on a challenging patient. At the very least, tape or sandbags may be required to immobilize the legs at the proper degree of internal rotation.

GERIATRIC APPLICATIONS

Geriatric patients are prone to hip fractures resulting from falls and an increased incidence of osteoporosis. As noted earlier, the position of the patient's foot and leg must be observed in trauma cases. **It is critical that the injured limb not be moved if the leg is externally rotated.** An AP projection of both hips for comparison should be taken first, without movement of the affected limb, to check for fractures. This step may be followed by an inferosuperior (Danelius-Miller method) projection of the affected hip.

In nontrauma situations, most geriatric patients require (and appreciate) some immobilization to assist them in holding their feet and legs inverted for the AP pelvis and to support the limb for the lateral projection.

Patients who have undergone hip replacement surgery should **not** be placed in the frog-leg position for any postsurgical procedures. An inferosuperior lateral is indicated in addition to the AP projection.

Fig. 7.33 Male gonadal shielding for hips and pelvis.

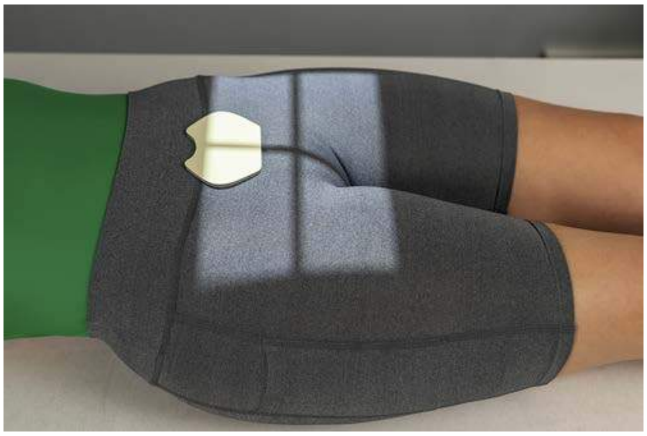


Fig. 7.34 Female gonadal (ovarian) shielding for bilateral hips and proximal femora.

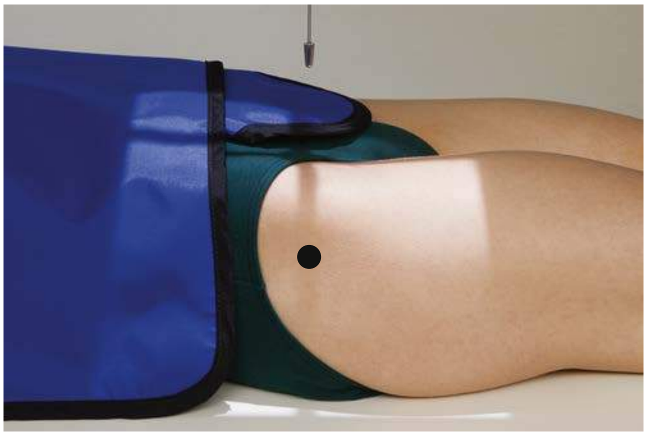


Fig. 7.35 General abdominal and pelvic shielding for proximal femur to include hip.

7

BARIATRIC PATIENT CONSIDERATIONS

Bariatric patients may present positioning challenges when imaging the pelvis, hips, and upper femurs. Increased adipose tissue adds subject density and may require an increase in technical factors. These changes may include an increase in kVp to improve penetration through additionally thick tissue. mA and time may also be increased, but sparingly to avoid excessive patient dose. The use of a grid for anatomic structures over 10 cm can be used to eliminate the demonstration of scatter.

Frequently used positioning landmarks may be difficult to palpate. Bony anatomy does not change unless major pathology, such as multiple fractures, has displaced the bones. Although the soft tissue may make it appear that the bones are larger or are farther apart, generally this is not the case. If common landmarks cannot be found, ask the patient to point out the ASIS, iliac crest, or the symphysis pubis on themselves. Additionally, bariatric patients may have difficulty holding oblique or lateral positions; the use of positioning sponges or other devices is recommended to ensure the patient's comfort and safety.

Digital Imaging Considerations

The following text provides a summary of the guidelines that should be followed with digital imaging of the procedures described in this chapter.

1. **Close collimation:** Collimating to the body part being imaged and **accurate centering** are most important when imaging the hip and pelvis.
2. **Exposure factors:** It is important that the ALARA principle (exposure to patient as low as reasonably achievable) be followed and the **lowest exposure factors required to obtain a diagnostic image be used.** This includes the highest kVp and lowest mAs that will result in desirable image quality.
3. **Post-processing evaluation of exposure indicator:** The exposure indicator on the final processed image must be checked to verify the exposure factors used were in the correct range **to ensure an optimum quality image with the least radiation dose to the patient.**
4. **Compensating filters:** The use of a compensating filter for axiolateral projections of the hip will allow better penetration of the femoral head while preventing overexposure of the femoral neck and shaft region.

Alternative Modalities

COMPUTED TOMOGRAPHY (CT)

CT is useful for evaluating soft tissue involvement of lesions or determining the extent of fractures. CT also is helpful for studying the relationship of the femoral head to the acetabulum before hip surgery or for performing a postreduction study of a developmental hip dislocation.

In general, CT is useful to add to the anatomic or pathologic information already obtained by conventional radiography. For children, the CT examination is useful for examining the relationship of the femoral head to the acetabulum after surgical reduction of a developmental hip dislocation.

Fractures of the pelvic ring missed on conventional radiographic projections, especially those involving the ischial and pubic rami, often are demonstrated during CT scanning.

MAGNETIC RESONANCE IMAGING (MRI)

Similar to CT, MRI can be useful for imaging the lower limb or pelvis when soft tissue injuries or possible abnormalities related to joints are suspected. In general, depending on clinical history, MRI may be used when additional information not obtained from conventional radiographs is needed.

SONOGRAPHY (ULTRASOUND)

Ultrasound is useful for evaluating newborns for hip dislocation and for assessing joint stability during movement of the lower limbs. This method usually is selected during the first 4 to 6 months of infancy to reduce ionizing radiation exposure.

NUCLEAR MEDICINE (NM)

NM scans can be useful in providing **early evidence** of certain bony pathologic processes, such as occult fractures, bone infections, metastatic carcinoma, or other metastatic or primary malignancies. NM is more sensitive and generally provides earlier evidence than other modalities because it assesses the physiologic aspect rather than the anatomic aspect of these conditions.

Clinical Indications

Clinical indications involving the pelvis and hips with which technologists should be familiar include the following (not necessarily an inclusive list):

Ankylosing spondylitis: The first effect demonstrated is fusion of the sacroiliac joints. The disease causes extensive calcification of the anterior longitudinal ligament of the spinal column. It is progressive, working up the vertebral column and creating a radiographic characteristic known as bamboo spine. Males are most often affected.

Avulsion fractures of the pelvis: These fractures cause extreme pain and are difficult to diagnose if not imaged properly. Fractures occur in adolescent athletes who experience sudden, forceful, or unbalanced contraction of the tendinous and muscular attachments, such as might occur while running hurdles. The force of the tendons and muscles sliding over the tuberosities, ASIS, anterior inferior iliac spine (AIIS), superior corner of the symphysis pubis, and iliac crest may cause avulsion fractures.[4]

Chondrosarcoma: A malignant tumor of the cartilage, it usually occurs in the pelvis and long bones of men older than 45 years. A chondrosarcoma may be completely removed surgically if it does not respond to radiation or chemotherapy.

Developmental dysplasia of the hip (DDH) (older term is *congenital dislocation of the hip* [CDH]): These hip dislocations are caused by conditions present at birth and may require frequent hip radiographs (see Chapter 16).

Legg-Calvé-Perthes disease: The most common type of aseptic or ischemic necrosis. Lesions typically involve only one hip (head and neck of femur). The disease occurs predominantly in 5- to 10-year-old boys, and a limp is usually the first clinical sign. Radiographs demonstrate a flattened femoral head that later can appear fragmented.

Metastatic carcinoma: The malignancy spreads to the bone via the circulatory system or lymphatic system, or by direct invasion. Metastatic tumors of the bone are much more common than primary malignancies. Bones that contain red bone marrow are the more common metastatic sites (spine, skull, ribs, pelvis, and femora).

Osteoarthritis: This condition is known as a *degenerative joint disease (DJD),* with degeneration of joint cartilage and adjacent bone causing pain and stiffness. It is the most common type of arthritis and may be considered a normal part of the aging process. It is common in weight-bearing joints such as the hips, and first evidence is seen on radiographic images of joints such as the hip before symptoms develop, in many persons by age 40. As the condition worsens, joints become less mobile, and new growths of cartilage and bone are seen as osteophytes (bony outgrowths).

Pelvic ring fractures: Because of the closed ring structure of the pelvis, a severe blow or trauma to one side of the pelvis may result in a fracture opposite from the site of primary trauma, thus requiring clear radiographic visualization of the entire pelvis. This type of trauma is referred to as a **contrecoup injury.**

Proximal femur (hip) fractures: These fractures are most common in older adult or geriatric patients with osteoporosis or avascular necrosis. Both osteoporosis (loss of bone mass from metabolic or other factors) and avascular (loss of blood circulation) necrosis (cell death) frequently lead to weakening or collapse of weight-bearing joints such as the hip joint; fractures occur with only minimal trauma.

Slipped capital femoral epiphysis (SCFE): This condition usually occurs in 10- to 16-year-olds during rapid growth, when even minor trauma can precipitate its development. The epiphysis appears shorter and the epiphyseal plate wider, with smaller margins.

Routine and Special Projections

Certain routine and special projections or positions for the proximal femora and pelvis are demonstrated and described on the following pages as suggested standard and special departmental procedures.

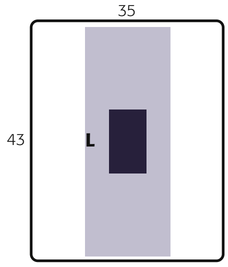
AP PROJECTION: FEMUR—MID AND DISTAL

NOTE: If the site of interest is in the area of the proximal femur, a unilateral hip routine or pelvis is recommended, as described in this chapter.

Femur—Mid and Distal
ROUTINE
- AP
- Lateral

Clinical Indications

- Mid- and distal femur, including knee joint, for detection and evaluation of fractures and/or bone lesions



Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—14 × 17 inches (35 × 43 cm), portrait
- Grid
- kVp range—75–85

Shielding Shield radiosensitive tissues outside the region of interest. Ensure that shielding does not obscure any aspect of the femur.

Patient Position Place patient in the supine position, with femur centered to midline of table; give pillow for head. (This projection also may be done on a stretcher with a portable grid placed under the femur.)

Part Position

- Align femur to CR and to midline of table or IR.
- Rotate leg internally approximately 5° for a true AP, as for an AP knee. (For proximal femur, 15° to 20° internal leg rotation is required, as for an AP hip.)
- Ensure that knee joint is included on IR, considering the divergence of the x-ray beam (Fig. 7.36). (Lower IR margin should be approximately 2 inches [5 cm] below knee joint.)

CR

- CR is **perpendicular** to femur and IR.
- Direct CR to **midpoint of IR.**

Recommended Collimation Collimate closely on both sides to femur with end collimation to film borders.

Including both Joints Common departmental routines include both joints on all initial femur exams. For a large adult, a second smaller IR then should be used for an AP of the knee or hip, ensuring that both hip and knee joints are included. If the hip is included, the leg should be rotated 15° to 20° internally to place the femoral neck in profile.

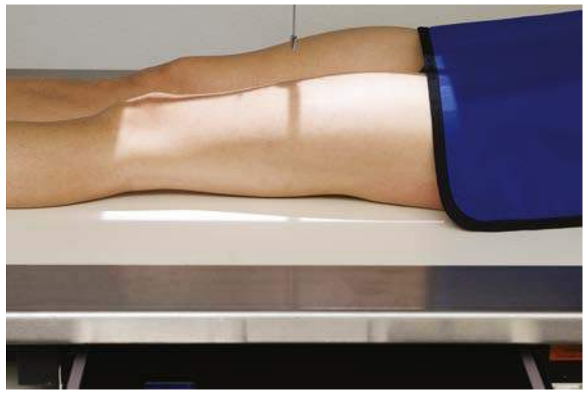


Fig. 7.36 AP—mid- and distal femur (head at cathode end).

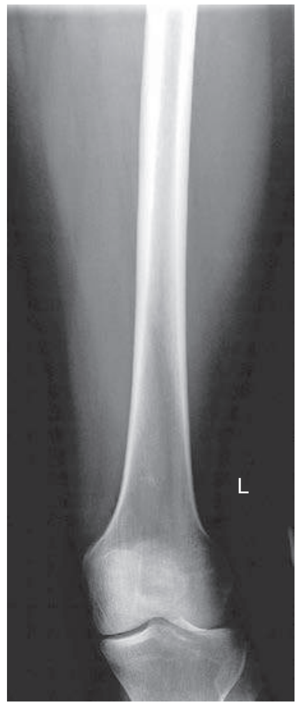


Fig. 7.37 AP—mid- and distal femur.

Evaluation Criteria

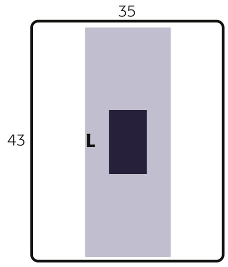
Anatomy Demonstrated: • Distal two-thirds of distal femur, including knee joint, is shown. • Knee joint space will not appear fully open because of divergent x-ray beam (Fig. 7.37).
Position: • **No rotation** is evidenced; femoral and tibial condyles should appear symmetric in size and shape with the outline of patella slightly toward medial side of femur. • The approximate medial half of fibular head should be superimposed by tibia. Femur should be centered to collimation field and aligned with long axis of IR with knee joint space a minimum of 1 inch (2.5 cm) from distal IR margin. • Collimation to **area of interest.**
Exposure: • Optimal exposure with correct use of anode heel effect or use of compensating filter will result in near uniform density (brightness) of entire femur. • **No motion** should occur; fine trabecular markings should be clear and sharp throughout length of femur.

LATERAL—MEDIOLATERAL OR LATEROMEDIAL PROJECTIONS: FEMUR—MID AND DISTAL

NOTE: For possible trauma, if the site of interest is in the area of the proximal femur, a unilateral trauma hip routine is recommended. For non-trauma, lateral of proximal femur (see p. 290).

Femur—Mid and Distal
ROUTINE
- AP
- Lateral

Clinical Indications
- Mid- and distal femur, including knee joint, for detection and evaluation of fractures and/or bone lesions



Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—14 × 17 inches (35 × 43 cm), portrait
- Grid
- kVp range—75–85

Shielding Shield radiosensitive tissues outside region of interest. Ensure shielding does not obscure any aspect of femur.

Patient Position Place patient in the lateral recumbent position, or supine for trauma patient.

Part Position

Lateral Recumbent (Fig. 7.38)
WARNING: Do not attempt this position if patient has severe trauma.

- Flex knee approximately 45° with patient on affected side, and align femur to midline of table or IR.
- Place unaffected leg behind affected leg to prevent over-rotation.
- Adjust IR to include knee joint (lower IR margin should be approximately 2 inches [5 cm] below knee joint). A second IR to include the proximal femur and hip generally will be required on an adult (see p. 290).

Trauma Lateromedial Projection (Fig. 7.39)
- Place support under affected leg and knee and support foot and ankle in true AP position.
- Place IR on edge against medial aspect of thigh to include knee, with horizontal x-ray beam directed from lateral side.

CR
- CR **perpendicular** to femur and directed to **midpoint of IR**

Recommended Collimation Collimate closely on both sides to femur with end collimation to IR borders.

Evaluation Criteria
Anatomy Demonstrated: • Distal two-thirds of distal femur, including the knee joint, is shown. • Knee joint will not appear open, and distal margins of the femoral condyles will not be superimposed because of divergent x-ray beam (Fig. 7.40).
Position, True Lateral: • Anterior and posterior margins of medial and lateral femoral condyles should be superimposed and aligned with open patellofemoral joint space. • Femur should be centered to collimation field with knee joint space a minimum of 1 inch (2.5 cm) from distal IR margin. • Collimation to **area of interest.**
Exposure: • Optimal exposure with correct use of anode heel effect or use of compensating filter will result in near-uniform density (brightness) of entire femur. • **No motion** is present; fine trabecular markings should be clear and sharp throughout length of femur.

Fig. 7.38 Mediolateral mid- and distal femur.

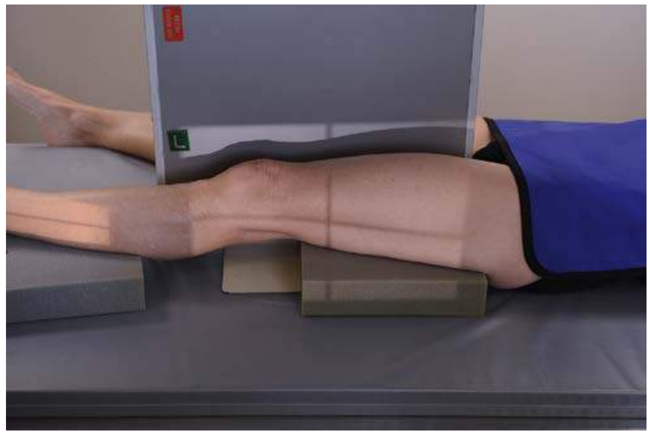


Fig. 7.39 Trauma lateromedial (horizontal beam) projection. Note: When a grid cassette is used, care must be taken to prevent grid cutoff.

Fig. 7.40 Lateral—mid- and distal femur. (From Fagan R, Furey AJ. Use of large osteochondral allografts in reconstruction of traumatic uncontained distal femoral defects. *Journal of Orthopaedics* 11(1): 43–47, 2014.)

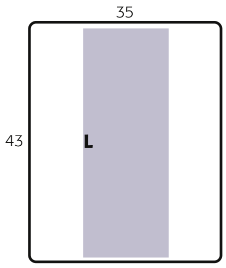
LATERAL—MEDIOLATERAL PROJECTION: FEMUR—MID AND PROXIMAL

WARNING: Do not attempt this position for patients with possible fracture of the hip or proximal femur. Refer to trauma lateral hip routine in this chapter.

Clinical Indications
- Mid- and proximal femur, including lateral hip, for detection and evaluation of fractures and bone lesions

Femur—Mid and Proximal
ROUTINE
• AP
• Lateral

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—14 × 17 inches (35 × 43 cm), portrait
- Grid
- kVp range—75–85



Shielding Shield radiosensitive tissues outside region of interest. Ensure shielding does not obscure any aspect of femur.

Patient Position Place patient in the lateral recumbent position, with affected side down; provide pillow for head.

Part Position
- Flex affected knee 45° and align femur to midline of table. (Remember the proximal and midportions of the femur are nearer to the anterior aspect of the thigh.)
- Extend and support unaffected leg behind affected knee and have patient roll back (posteriorly) 15° to prevent superimposition of proximal femur and hip joint (Fig. 7.41).
- Adjust IR to include hip joint, considering the divergence of the x-ray beam. (Palpate ASIS and place upper IR margin at the level of this landmark.)

CR
- CR **perpendicular** to femur
- CR directed to **midpoint of IR**

Recommended Collimation Collimate closely on both sides of femur with end collimation to IR borders.

NOTE: Alternate routine to include both joints: Common departmental routines include both joints on all initial femur examinations. On a large adult, this requires a second smaller IR (10 x 12 inches [24 x 30 cm]) of the hip or knee joint.

Evaluation Criteria

Anatomy Demonstrated: • Proximal one-half to two-thirds of the proximal femur, including the hip joint, is shown. • Proximal femur and hip joint should not be superimposed by opposite limb (Figs. 7.42 and 7.43).
Position, True Lateral: • Superimposition of the greater and lesser trochanters by the femur exists, with only a small part of the trochanters visible on medial side. • Most of the greater trochanter should be superimposed by the neck of the femur. Femur should be centered to collimation field with hip joint a minimum of 1 inch (2.5 cm) from proximal IR margin. • Collimation to **area of interest.**
Exposure: • Optimal exposure with correct use of anode heel effect or use of compensating filter will result in near-uniform density (brightness) of entire femur. • **No motion** is present; fine trabecular markings should be clear and sharp throughout length of femur.

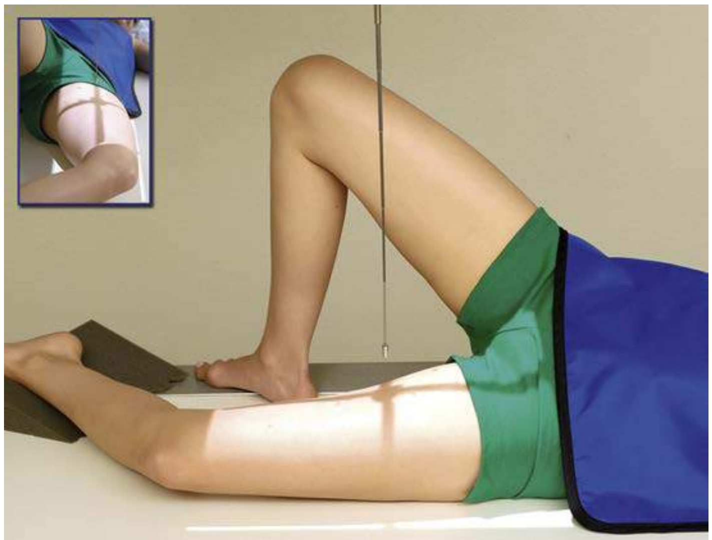


Fig. 7.41 Mediolateral—mid- and proximal femur.

Fig. 7.42 Mediolateral—mid- and proximal femur.

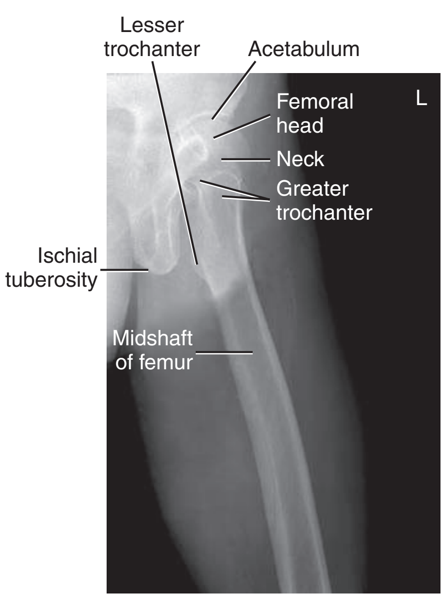


Fig. 7.43 Mediolateral—mid- and proximal femur.

AP PELVIS PROJECTION (BILATERAL HIPS): PELVIS

WARNING: Do not attempt to rotate legs internally if a hip fracture or dislocation is suspected. Perform position with minimal movement of affected leg.

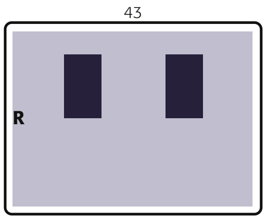
Pelvis
ROUTINE
- **AP**

Clinical Indications

- Fractures, joint dislocations, degenerative disease, and bone lesions

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—14 × 17 inches (35 × 43 cm), landscape
- Grid
- kVp range—80–90



Shielding Shield radiosensitive tissues outside the region of interest. Shield gonads on all male patients. Ovarian shielding on females, however, generally is not possible without obscuring essential pelvis anatomy (unless interest is in area of hips only).

Patient Position With patient supine, place arms at sides or across upper chest; provide pillow for head and support under knees; may be done erect with correction of lower limbs to rotate proximal femora into anatomic position and if **no fracture is suspected.** Position can also be performed erect.

Part Position

- Align midsagittal plane of patient to centerline of table and CR.
- Ensure that pelvis is **not rotated;** distance from tabletop to each ASIS should be equal.
- Separate legs and feet, then **internally rotate** long axes of feet and entire lower limb **15° to 20°** (see WARNING). Technologist may have to place sandbag between heels and tape top of feet together or use additional sandbags against feet to retain this position (Fig. 7.44, insert). Erect positioned similar to recumbent version (Fig. 7.44)

CR

- CR is perpendicular to IR, directed midway between level of ASIS and the symphysis pubis. This is approximately 2 inches (5 cm) inferior to level of ASIS (see NOTE).
- Center CR to IR.

Recommended Collimation Collimate on four sides to anatomy of interest.

Evaluation Criteria

Anatomy Demonstrated: • Pelvic girdle, L5, sacrum and coccyx, femoral heads and neck, and greater trochanters are visible (Figs. 7.45 and 7.46).

Position: • Lesser trochanters should not be visible at all; for many patients, only the tips are visible. Greater trochanters should appear equal in size and shape. • **No rotation** is evidenced by symmetric appearance of the iliac alae, or wings, the ischial spines, and the two obturator foramina. A foreshortened or closed obturator foramen indicates rotation in that direction. (A closed or narrowed right obturator foramen compared with the left indicates rotation toward the right.) • The right and left ischial spines (if visible) should appear equal in size. Correct centering evidenced by demonstration of entire pelvis and superior femora without foreshortening in collimated field. • Collimation to **area of interest.**

Exposure: • Optimal exposure visualizes L5 and sacrum area and margins of the femoral heads and acetabula, as seen through overlying pelvic structures, without overexposing the ischium and pubic bones. • Trabecular markings of proximal femora and pelvic structures appear sharp, indicating **no motion.**

Respiration Suspend respiration during exposure.

NOTE: If performed as part of a hip routine, centering should be 2 inches (5 cm) lower to level of midfemoral heads or necks to include more of proximal femora.

Fig. 7.44 Patient and part position—AP pelvis.

Fig. 7.45 AP pelvis.

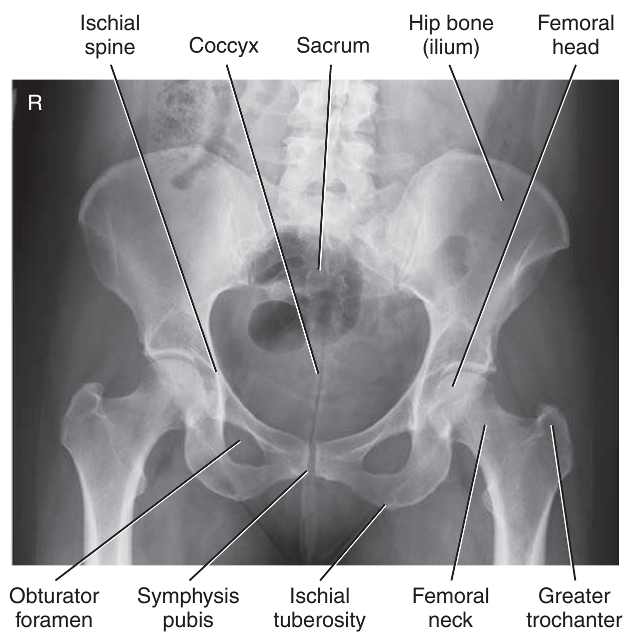


Fig. 7.46 AP pelvis.

7

AP BILATERAL FROG-LEG PROJECTION: PELVIS

MODIFIED CLEAVES METHOD

WARNING: Do not attempt this position on a patient with destructive hip disease or with potential hip fracture or dislocation.

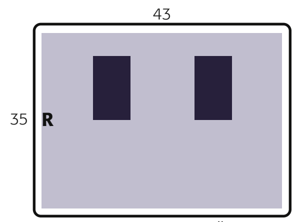
Pelvis
ROUTINE
- AP
- AP bilateral frog-leg (modified Cleaves method)

Clinical Indications

- Demonstration of a nontrauma hip
- Developmental dysplasia of hip (DDH), also known as congenital hip dislocation (CHD)

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—14 × 17 inches (35 × 43 cm), landscape
- Grid
- kVp range—80–90



No center AEC cell with shield in place.

Shielding Shield radiosensitive tissues outside the region of interest. Shield gonads for both males and females without obscuring essential anatomy (see NOTE 1).

Patient Position With patient supine, provide pillow for head and place arms across chest.

Part Position

- Align patient to midline of table and/or IR and to CR.
- Ensure pelvis is **not rotated** (equal distance of ASIS to tabletop).
- Center IR to CR, at level of femoral heads, with top of IR approximately at level of iliac crest.
- Flex both knees approximately 90°, as demonstrated.
- Place the plantar surfaces of feet together and **abduct both femora 40° to 45° from vertical** (see NOTE 2). Ensure that **both femora are abducted the same amount** and that pelvis is **not rotated** (Fig. 7.47).
- Place supports under each leg for stabilization if needed.

CR

- CR is **perpendicular** to IR, directed to a point **3 inches (7.5 cm) below level of ASIS** (1 inch [2.5 cm] above symphysis pubis).

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration during exposure.

NOTE 1: This projection frequently is performed for periodic follow-up examinations on younger patients. Correct placement of gonadal shielding is important for male and female patients, ensuring that hip joints are not obscured (Fig. 7.48).

NOTE 2: Less abduction of femora of only 20° to 30° from vertical plane provides for the least foreshortening of femoral head and neck region, but this placement foreshortens the entire proximal femora, which may not be desirable.

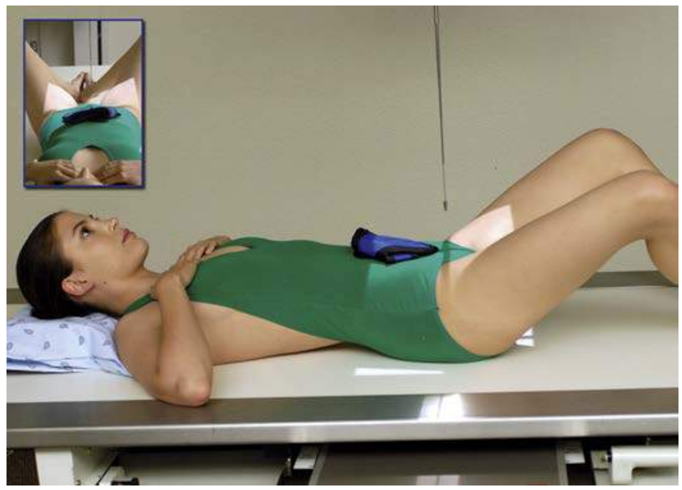


Fig. 7.47 Bilateral frog-leg—femora abducted 40° to 45°.

Evaluation Criteria

Anatomy Demonstrated: • Femoral heads and necks, acetabulum, and trochanteric areas are visible on one radiograph (see Fig. 7.48).

Position: • **No rotation,** as evidenced by symmetric appearance of the pelvic bones, especially the ala of the ilium, two obturator foramina, and ischial spines, if visible. • The femoral heads and necks and greater and lesser trochanters should appear symmetric if both thighs were abducted equally. • The lesser trochanters should appear equal in size, as projected beyond the lower or medial margin of the femora. • Most of the area of the greater trochanters appears superimposed over the femoral necks, which appear foreshortened (see NOTE 2). • Collimation to **area of interest.**

Exposure: • Optimal exposure visualizes the margins of the femoral head and the acetabulum through overlying pelvic structures, without overexposing the proximal femora. • Trabecular markings appear sharp, indicating **no motion.**

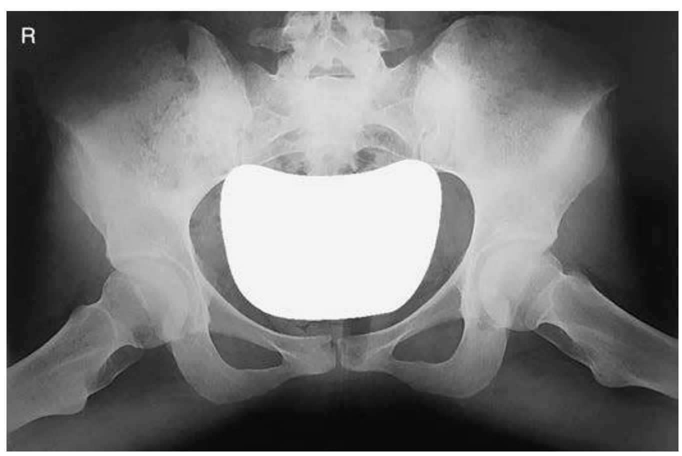


Fig. 7.48 Bilateral frog-leg (ovarian shield in place). (Courtesy Kathy Martensen, BS, RT[R].)

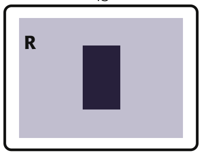
AP AXIAL OUTLET PROJECTION[5] (FOR ANTERIOR-INFERIOR PELVIC BONES): PELVIS

TAYLOR METHOD

Pelvis
SPECIAL
- AP axial outlet projection

Clinical Indications
- Bilateral view of the bilateral pubis and ischium to allow assessment of pelvic trauma for fractures and displacement

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—14 × 17 inches (35 × 43 cm), landscape
- Grid
- kVp range—80–90

43
35


Shielding Shield radiosensitive tissues outside region of interest.

Patient Position With patient supine, provide pillow for head. With patient's legs extended, place support under knees for comfort (Fig. 7.49).

Part Position
- Align midsagittal plane to CR and to midline of table and/or IR.
- Ensure **no rotation** of pelvis (ASIS-to-tabletop distance equal on both sides).
- Center IR to projected CR.

CR
- Angle CR **cephalad 20° to 35° for males** and **30° to 45° for females.** (These different angles are caused by differences in the shape of male and female pelvises. See section *Male Versus Female Pelvis Differences* on pg. 270.)
- Direct CR to a midline point 1 to 2 inches (2.5 to 5 cm) distal to the superior border of the symphysis pubis or greater trochanters.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration during exposure.

Evaluation Criteria
Anatomy Demonstrated: • Superior and inferior rami of pubis and body and ramus of ischium are demonstrated well, with minimal foreshortening or superimposition (Figs. 7.50 and 7.51).
Position: • **No rotation:** Obturator foramina and bilateral ischia are equal in size and shape. Correct CR angle evidenced by demonstration of the anterior/inferior pelvic bones, with minimal foreshortening. Midpoint of symphysis joint should be at center of collimated field. • Collimation to **area of interest.**
Exposure: • Body and superior rami of pubis are well demonstrated without overexposure of ischial rami. • Bony margins and trabecular markings of pubic and ischial bones appear sharp, indicating **no motion.**

Fig. 7.49 AP axial outlet projection—CR 40° cephalad.

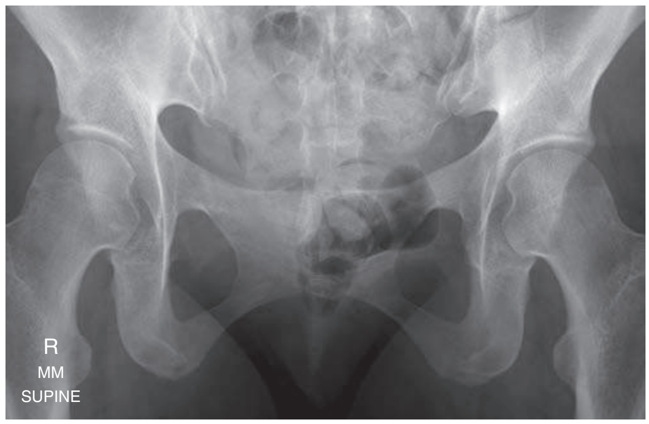


Fig. 7.50 AP axial outlet projection. (Courtesy Joss Wertz, DO.)

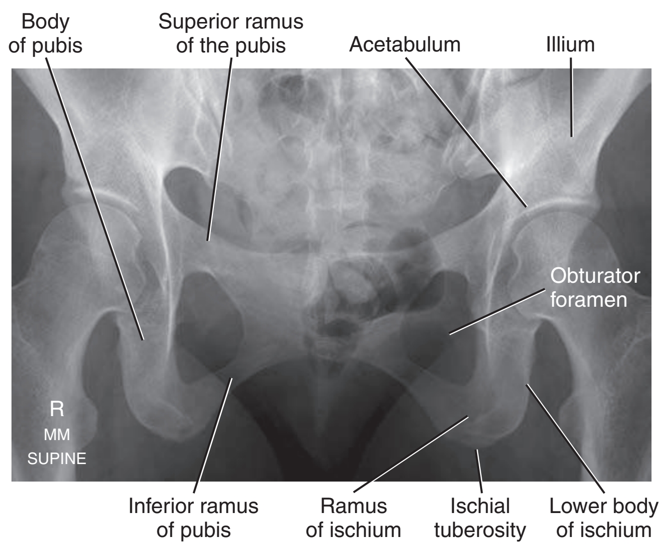


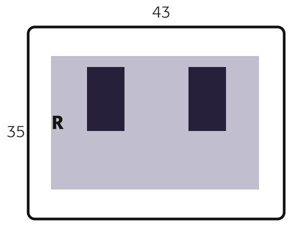
Fig. 7.51 AP axial outlet projection. (Courtesy Joss Wertz, DO.)

AP AXIAL INLET PROJECTION[5]: PELVIS

Clinical Indications
- Assessment of pelvic trauma for posterior displacement or inward or outward rotation of the anterior pelvis

Pelvis
SPECIAL
- AP axial outlet projection
- AP axial inlet projection

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—14 × 17 inches (35 × 43 cm), landscape
- Grid
- kVp range—80–90



Shielding Shield radiosensitive tissues outside region of interest. Gonadal shielding is possible for males if care is taken not to obscure essential pelvic anatomy.

Patient Position
With patient supine, provide pillow for head. With patient's legs extended, place support under knees for comfort (Fig. 7.52).

Part Position
- Align midsagittal plane to CR and to midline of table and/or IR.
- Ensure **no rotation** of pelvis (ASIS-to-tabletop distance equal on both sides).
- Center IR to projected CR.

CR
- Angle CR **caudad 40°** (near perpendicular to plane of inlet).
- Direct CR to a midline point at level of ASIS.

Recommended Collimation Collimate closely on four sides to area of interest.

Respiration Suspend respiration during exposure.

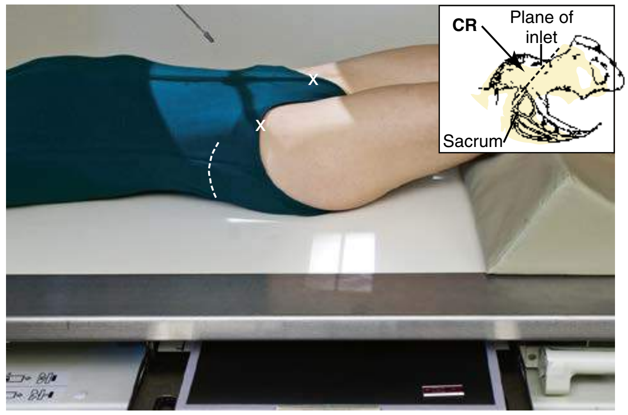


Fig. 7.52 AP axial inlet projection—CR 40° caudad (CR perpendicular to pelvic inlet).

Fig. 7.53 AP axial inlet projection.

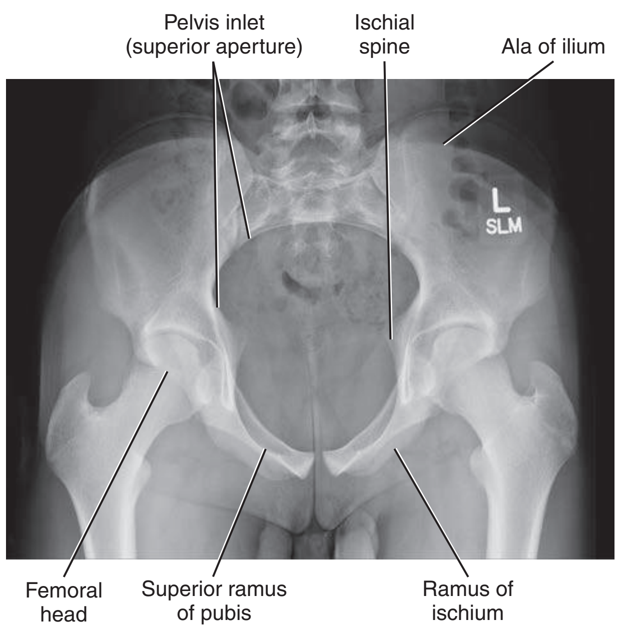


Fig. 7.54 AP axial inlet projection.

Evaluation Criteria

Anatomy Demonstrated: • Axial projection that demonstrates the pelvic ring or inlet (superior aperture) in its entirety (Figs. 7.53 and 7.54)

Position: • **No rotation:** Ischial spines are fully demonstrated and equal in size and shape. Proper centering and angulation are evidenced by demonstration of the superimposed anterior and posterior portions of the pelvic ring. • Center of pelvic inlet should be at center of collimated field. • Collimation to **area of interest.**

Exposure: • Optimal exposure demonstrates the superimposed anterior and posterior portions of the pelvic ring. Lateral aspects of ala generally are overexposed. • Bony margins and trabecular markings of pubic and ischial bones appear sharp, indicating **no motion.**

POSTERIOR OBLIQUE PROJECTION: PELVIS—ACETABULUM

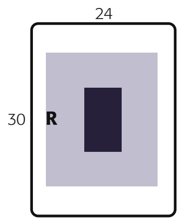
JUDET METHOD

Clinical Indications

- Acetabular fracture
- Pelvic ring fractures

Right and **left oblique projections** generally are taken for comparison, with both centered for upside or both for downside acetabulum. Possible pelvic ring fractures due to a contrecoup injury, the entire pelvis must be included. In this case, centering should be adjusted to include both hips.

Pelvis
SPECIAL
- AP axial outlet
- AP axial inlet
- Posterior oblique acetabulum (Judet method)
- Posterior axial oblique acetabulum (Teufel method)



Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait, or 14 × 17 (35 × 43 cm), landscape, if both hips must be seen on each projection
- Grid
- kVp range—80–90

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position—Posterior Oblique Positions

- With patient semisupine, provide pillow for head and position for **affected side up or down,** depending on anatomy to be demonstrated.

Part Position

- Place patient in **45° posterior oblique,** with both pelvis and thorax 45° from tabletop. Support with wedge sponge.
- Align femoral head and acetabulum of interest to midline of tabletop and/or IR.
- Center IR longitudinally to CR at level of femoral head.

CR

Acetabulum

- Affected side **down:** direct CR perpendicular and centered to **2 inches** (5 cm) **distal and 2 inches** (5 cm) **medial to downside ASIS** (Fig. 7.55, insert).
- Affected side **up:** direct perpendicular and centered to **2 inches** (5 cm) **directly distal to upside ASIS** (Fig. 7.56, insert).

Pelvic Ring

- Direct CR perpendicular and centered to 2 inches (5 cm) inferior from level of ASIS and 2 inches (5 cm) medial to upside ASIS. (Figs. 7.55 and 7.56)

Recommended Collimation

- 10 × 12 inches (24 × 30 cm) or 14 × 17 inches (35 × 43 cm), depending on size of IR selected and field of view; or, collimate on four sides to anatomy of interest

Respiration Suspend respiration for the exposure.

7

Fig. 7.55 RPO—centered for right (downside) acetabulum.

Fig. 7.56 LPO—centered for right (upside) acetabulum.

Evaluation Criteria

Anatomy Demonstrated:
Acetabulum: • When centered to the **downside** acetabulum, the **anterior rim** of the acetabulum and the **posterior (ilioischial) column** are demonstrated. The **iliac wing** also is well visualized (Figs. 7.57 and 7.58). • When centered to the **upside** acetabulum, the **posterior rim** of the acetabulum and the **anterior (iliopubic) column** are demonstrated. The **obturator foramen** also is visualized (Figs. 7.59 and 7.60).
Pelvic Ring: • Projections will demonstrate the ilioischial and iliopubic columns, along with other aspect of pelvic ring (Figs. 7.61 and 7.62).
Position: • Proper degree of obliquity is evidenced by an open and uniform hip joint space at the rim of acetabulum femoral head. • The obturator foramen should be open, if rotated correctly, for the upside oblique, and should appear closed on downside oblique. Acetabulum (or pelvis) should be centered to IR and to collimation field. • Collimation to **area of interest.**
Exposure: • Optimal exposure should clearly demonstrate bony margins and trabecular markings of the acetabulum and femoral head regions; such markings should appear sharp, indicating **no motion.**

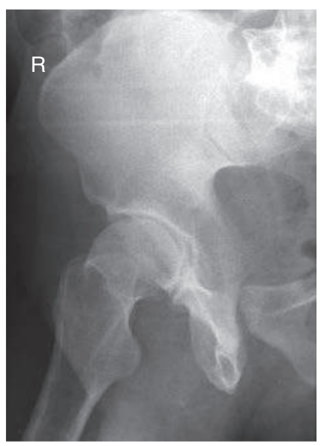


Fig. 7.57 RPO—downside (anterior rim and posterior [ilioischial] column).

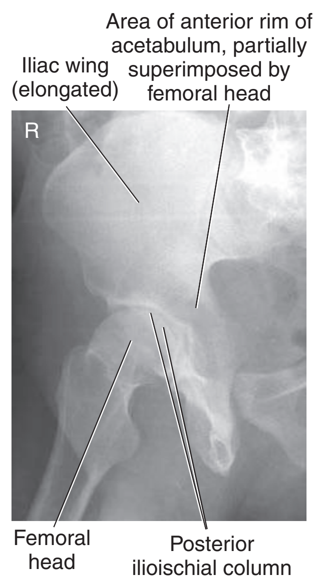


Fig. 7.58 RPO—downside acetabulum.

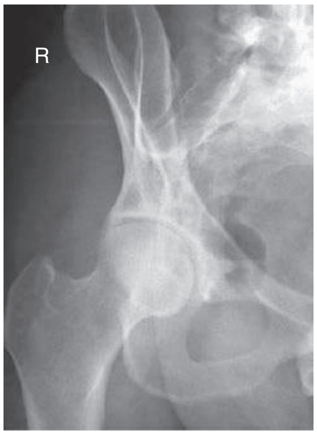


Fig. 7.59 LPO—upside (posterior rim and anterior [iliopubic] column.)

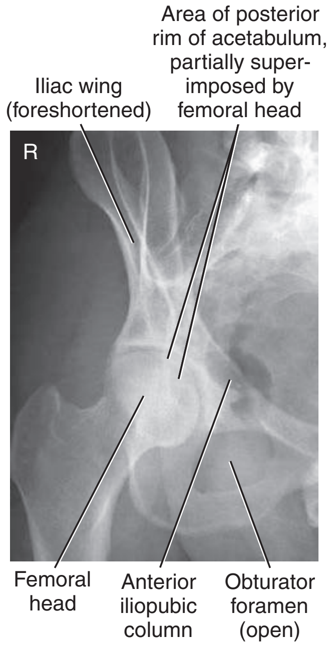


Fig. 7.60 LPO—upside acetabulum.

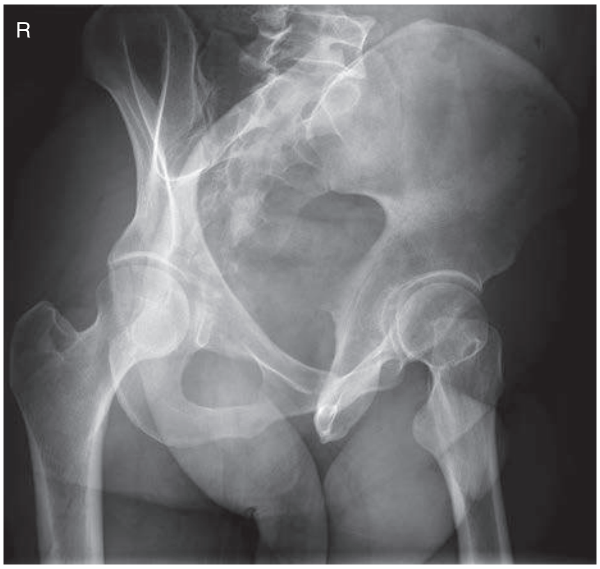


Fig. 7.61 LPO—full bilateral Judet method. (Case courtesy Dr Luke Danaher, Radiopaedia.org, rID: 39777.)

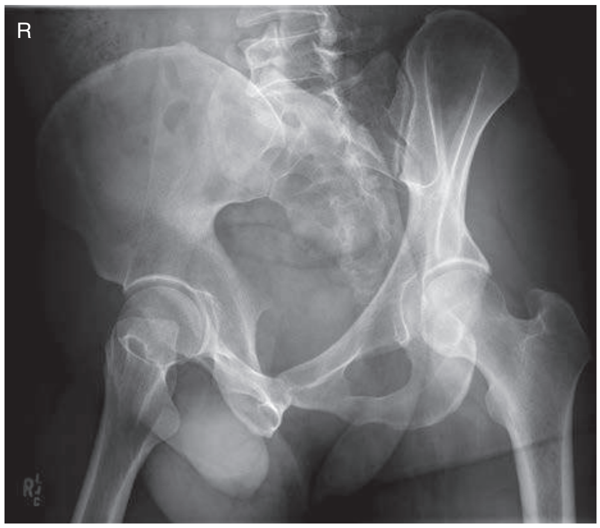


Fig. 7.62 RPO—Full bilateral Judet method. (Case courtesy Dr Luke Danaher, Radiopaedia.org, rID: 39777.)

PA AXIAL OBLIQUE PROJECTION: ACETABULUM

TEUFEL METHOD

Pelvis

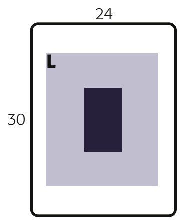
SPECIAL

- Acetabulum and femoral head, including the fovea capitis
- Posterior axial oblique acetabulum (Teufel method)

Clinical Indications

- Acetabular fracture, especially the superoposterior wall of the acetabulum

Right or left posterior **oblique** is taken to demonstrate the side of interest, centered to the downside acetabulum to demonstrate the hip joint and acetabulum in the center of the image, with the femoral head in profile. The concave area of the fovea capitis should be demonstrated, along with the superoposterior wall of the acetabulum.



Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range—75–85

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position—Axial Oblique Positions

- With patient semiprone, provide pillow for head and position for **affected side down.** Position can be performed erect

Part Position

- Place patient in **anterior oblique,** with both pelvis and thorax **35° to 40°** from tabletop or wall bucky. Support with wedge sponge (Fig. 7.63).
- Align femoral head and acetabulum of interest to midline of tabletop and/or IR.
- Center IR longitudinally to CR at level of femoral head.

CR

- When anatomy of interest is **downside,** direct CR perpendicular and centered to **1 inch** (2.5 cm) **superior to the level of the greater trochanter, approximately 2 inches** (5 cm) lateral to the midsagittal plane.
- Angle CR **12° cephalad.**

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration for the exposure.

Evaluation Criteria

Anatomy Demonstrated: • Centered to the **downside** acetabulum, the **superoposterior wall** of the acetabulum is demonstrated (Figs. 7.64 and 7.65).

Position: • Proper degree of obliquity is evidenced by visualization of the concave area of the fovea capitis with the femoral head in profile. • The obturator foramen should be open, if rotated correctly. Acetabulum should be centered to IR and to collimation field. • Collimation to **area of interest.**

Exposure: • Optimal exposure should clearly demonstrate bony margins and trabecular markings of the acetabulum and femoral head regions; such markings should appear sharp, indicating **no motion.**

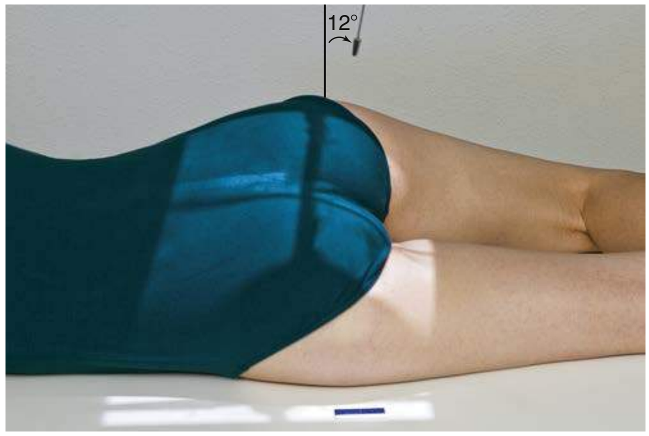


Fig. 7.63 PA axial oblique (Teufel) projection—12° cephalad angle.

Fig. 7.64 PA axial oblique (Teufel) projection.

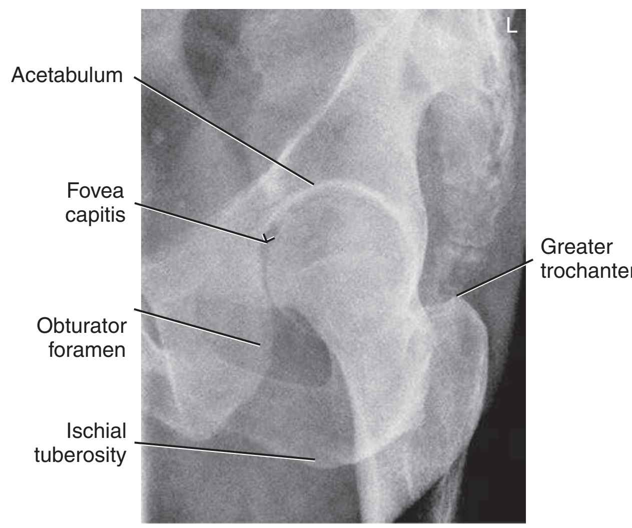


Fig. 7.65 PA axial oblique projection.

AP UNILATERAL HIP PROJECTION: HIP AND PROXIMAL FEMUR

WARNING: Do not attempt to rotate legs if fracture is suspected. An AP pelvis projection to include both hips for comparison should be completed before an AP unilateral hip is performed for possible hip or pelvis trauma.

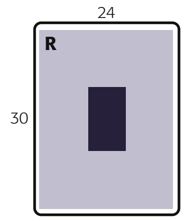
Clinical Indications

- **Postoperative** or **follow-up** examination to demonstrate the acetabulum, femoral head, neck, and greater trochanter
- Evaluation of condition and placement of any existing orthopedic appliance

Hip and Proximal Femur
ROUTINE
- AP unilateral hip
- Axiolateral (trauma hip) (inferosuperior)

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range—80–85



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position With patient supine, place arms at sides or across superior chest.

Part Position

- Locate **femoral neck** and align to CR and to midline of table and/or IR.
- Ensure **no rotation** of pelvis (equal distance from ASISs to table).
- Rotate affected leg **internally 15° to 20°** (see WARNING).

CR

- CR is perpendicular to the femoral neck (see p. 273 for femoral head and neck localization methods). Femoral neck can also be located about 1 to 2 inches (2.5 to 5 cm) medial and 3 to 4 inches (8 to 10 cm) distal to ASIS (Fig. 7.66).

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration during exposure.

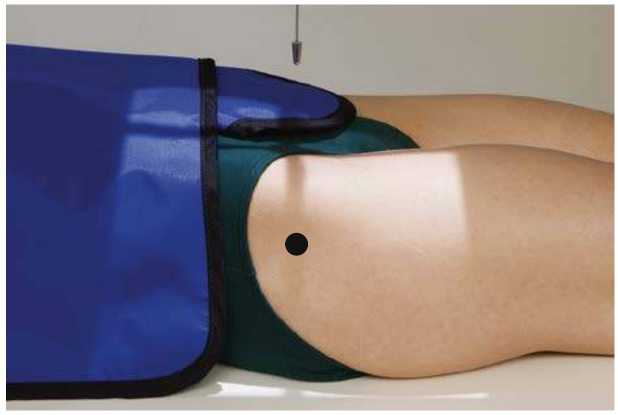


Fig. 7.66 AP right hip.

Fig. 7.67 AP hip.

Fig. 7.68 AP hip. (Copyright Getty Images/DieterMeyrl.)

Evaluation Criteria

Anatomy Demonstrated: • The proximal one-third of the femur should be visualized, along with the acetabulum and adjacent parts of the pubis, ischium, and ilium (Fig. 7.67). • Any existing orthopedic appliance should be visible in its entirety (Fig. 7.68).

Position: • The greater trochanter and femoral head and neck should be in full profile without foreshortening. • The lesser trochanter should not project beyond the medial border of the femur; with some patients, only the medial edge of it is seen with sufficient internal rotation of leg. Collimated field should demonstrate the entire hip joint and any orthopedic appliance in its entirety. • Collimation to **area of interest.**

Exposure: • Optimal exposure visualizes the margins of the femoral head and the acetabulum through overlying pelvic structures without overexposing other parts of the proximal femur or pelvic structures. • Trabecular markings of the greater trochanter and neck area appear sharp, indicating **no motion.**

AXIOLATERAL INFEROSUPERIOR PROJECTION—TRAUMA: HIP AND PROXIMAL FEMUR

DANELIUS-MILLER METHOD

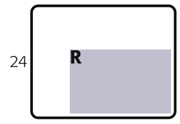
WARNING: Do not attempt to rotate leg internally on initial trauma examination.

NOTE: This is a common projection for trauma, surgery, and postsurgery patients, in addition to patients who cannot move or rotate the affected leg for frog-leg lateral.

Hip and Proximal Femur
ROUTINE
- AP unilateral hip
- Axiolateral (trauma) (inferosuperior)

Clinical Indications

- Lateral view for fracture or dislocation assessment in trauma hip situations when affected leg cannot be moved



Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), landscape **(portrait to long axis of femur)**
- Grid (grid perpendicular to the CR to prevent grid cutoff)
- kVp range—80–95

Shielding Shield radiosensitive tissues outside region of interest. Gonadal shielding, if used, must be carefully placed to not obscure any anatomy of the affected hip. Close collimation is important for reducing patient dose and improving image quality.

Patient Position May be done on stretcher or bedside if patient cannot be moved (see Chapter 15). Patient is supine, with pillow provided for head; elevate pelvis 1 to 2 inches (2.5 to 5 cm) if possible by placing supports under pelvis (more important for thin patients and for patients on a soft pad or in a bed).

Part Position (Figs. 7.69 and 7.70)

- Flex and elevate unaffected leg so that thigh is near vertical position and outside collimation field. Support in this position. (**Do NOT** place leg on collimator or x-ray tube due to risk of burns or electrical shock.)
- Check to ensure **no rotation** of pelvis (equal ASIS-table distance).
- Use hip localization method to identify location and alignment of femoral neck.
- Place IR in the crease above iliac crest and adjust so that it is **parallel to femoral neck** and **perpendicular to CR** (see Fig. 7.18). Use cassette holder if available, or use sandbags to hold image receptor/grid in place.
- Internally rotate affected leg 15° to 20° **unless contraindicated** by possible fracture or other pathologic process (see warning above).

CR CR is **perpendicular** to femoral neck and to IR.

Evaluation Criteria

Anatomy Demonstrated: • Entire femoral head and neck, trochanter, and acetabulum should be visualized along with any orthopedic prosthetic device in its entirety (Fig. 7.71).
Position: • Only a small part, if any, of lesser trochanter is visualized with inversion of affected leg. • Only the most distal part of femoral neck should be superimposed by greater trochanter. • Soft tissue from raised unaffected leg is not superimposed over affected hip if leg is raised sufficiently and CR is placed correctly. No grid lines are visible (grid lines indicate incorrect tube/IR alignment). • Collimation to **area of interest.**
Exposure: • Optimal exposure visualizes outline of entire femoral head and acetabulum without overexposing neck and proximal femoral shaft.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration during exposure.

NOTE: Demonstrating the most proximal portion of the femoral head and acetabulum on a patient with thick thighs may be challenging.

Fig. 7.69 Axiolateral hip. *Inset,* CR perpendicular and centered to femoral neck.

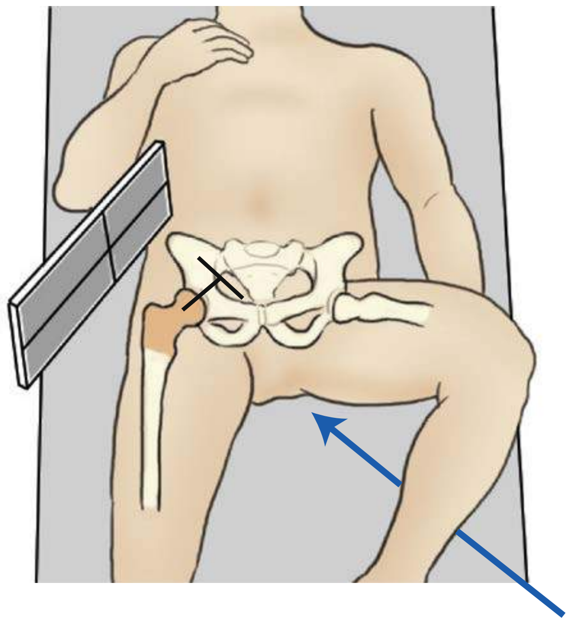


Fig. 7.70 Axiolateral hip projection.

Fig. 7.71 Axiolateral hip projection.

UNILATERAL FROG-LEG PROJECTION—MEDIOLATERAL: HIP AND PROXIMAL FEMUR

MODIFIED CLEAVES METHOD

WARNING: Do not attempt this position on patient with destructive hip disease or potential hip fracture or dislocation. This could result in significant displacement of fracture fragments (see lateral trauma projections).

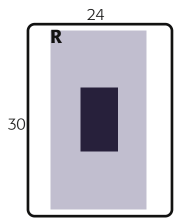
Clinical Indications

- Lateral view to assess hip joint and proximal femur for **nontraumatic hip** situations

Hip and Proximal Femur
SPECIAL—NONTRAUMA
• Unilateral frog-leg

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range—80–85



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position With patient erect or supine, position affected hip area to be aligned to CR and midline of table and/or IR.

Part Position (Fig. 7.72)

- Flex knee and hip on affected side, as shown, with sole of foot against inside of opposite leg, near knee if possible.
- Abduct femur **45° from vertical** for general proximal femur region (see NOTE 1).
- Center affected femoral neck to CR and midline of IR and tabletop. Apply hip localization methods to determine location of femoral neck.
- CR is **perpendicular** to IR (see NOTE 2), directed to **midfemoral neck** (center of IR).

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration during exposure.

NOTE 1: The optimum femur abduction for demonstration of the femoral neck with minimal distortion is **20° to 30°** from vertical on most patients. This results in significant foreshortening of the proximal femur region, which may be objectionable.

NOTE 2: A modification of this position is the Lauenstein-Hickey method, with the patient starting in a similar position, then rotating onto the affected side until the femur is in contact with the tabletop and parallel to the IR. This position foreshortens the neck region, but may demonstrate the head and acetabulum well if affected leg can be abducted sufficiently, as shown in Fig. 7.72, inset).

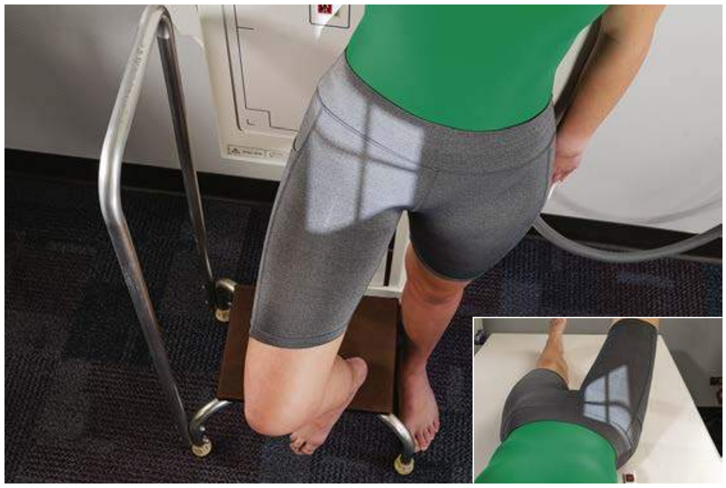


Fig. 7.72 45° abduction. Head and acetabulum are well demonstrated. *Inset,* 90° abduction. Unilateral frog-leg position (femoral neck parallel to IR). Femoral neck is foreshortened.

Fig. 7.73 For femoral neck—45° abduction.

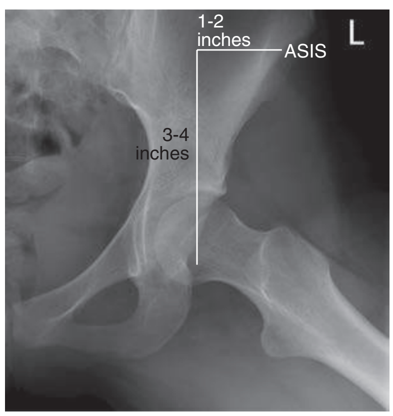


Fig. 7.74 Unilateral frog-leg, 20° to 30° abduction. (From McQuillen Martensen K. *Radiographic image analysis,* ed 4, St. Louis, 2015, Saunders Elsevier.)

Evaluation Criteria

Anatomy Demonstrated: • Lateral views of acetabulum and femoral head and neck, trochanteric area, and proximal one-third of femur are visible.

Position: • Proper abduction (45°) of femur is demonstrated by femoral neck seen in profile, superimposed by greater trochanter (Fig. 7.73). Less abduction (20-30°) will prevent superimposition of greater trochanter on the femoral neck (Fig. 7.74). Proper centering is evidenced by femoral neck at center of collimated field. • Collimation to **area of interest.**

Exposure: • Optimal exposure visualizes the margins of the femoral head and the acetabulum through overlying pelvic structures without overexposing other parts of the proximal femur. • Trabecular markings and bony margins of proximal femur and pelvis should appear sharp, indicating **no motion.**

MODIFIED AXIOLATERAL PROJECTION—POSSIBLE TRAUMA: HIP AND PROXIMAL FEMUR

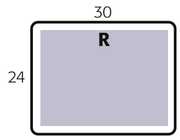
CLEMENTS-NAKAYAMA METHOD[6]

Clinical Indications

- Lateral oblique view is useful for assessment of possible **hip fracture** or with **arthroplasty** (surgery for hip prosthesis) when the patient has limited movement in both lower limbs and the inferosuperior projection cannot be obtained

Hip and Proximal Femur
SPECIAL—NONTRAUMA
• Unilateral frog-leg
SPECIAL—TRAUMA
• Modified axiolateral (Clements-Nakayama method)

Technical Factors



- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), landscape (LW to long axis of femur)
- Grid (IR on edge with 15° tilt; grid lines parallel to CR angle.)
- kVp range—80–90

Shielding Shield radiosensitive tissues outside region of interest without obscuring essential anatomy.

Patient Position With patient supine, position affected side near edge of table with both legs fully extended. Provide pillow for head and place arms across superior chest.

Part Position

- Maintain leg in neutral (anatomic) position (15° posterior CR angle compensates for internal leg rotation).
- Rest IR on extended bucky tray, which places the bottom edge of the IR approximately 2 inches (5 cm) below the level of the tabletop (Fig. 7.75).
- Tilt IR about 15° from vertical and adjust alignment of IR to ensure that face of IR is **perpendicular** to CR to prevent grid cutoff (Fig. 7.75, inset).
- Center centerline of IR to projected CR.

CR Angle CR mediolaterally as needed so that it is perpendicular to and centered to femoral neck. It should be angled posteriorly 15° to 20° from horizontal.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration during exposure.

Evaluation Criteria

Anatomy Demonstrated: • Lateral oblique views of acetabulum, femoral head and neck, and trochanteric area are visible (Figs. 7.76 and 7.77).

Position: • Femoral head and neck should be seen in profile, with only minimal superimposition by greater trochanter. • Lesser trochanter is seen projecting posterior to femoral shaft. (With leg in neutral or anatomic position, the amount of lesser trochanter seen is minimal, and with increased external rotation of leg, this amount decreases.) Femoral neck and trochanters should be centered to the image. • Collimation to **area of interest.**

Exposure: • Optimal exposure visualizes femoral head and neck without overexposing proximal femoral shaft. • No excessive grid lines are visible on radiograph. • Bony margins and trabecular markings should be visible and sharp, indicating **no motion.**

Fig. 7.75 Modified axiolateral—CR 15° tilt from horizontal perpendicular to femoral neck.

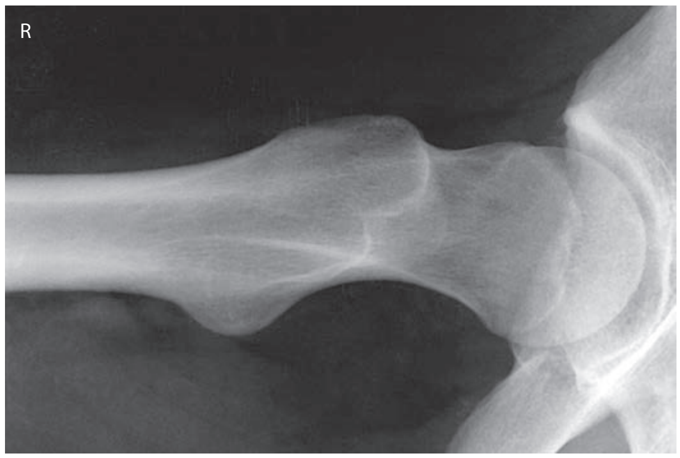


Fig. 7.76 Modified axiolateral projection.

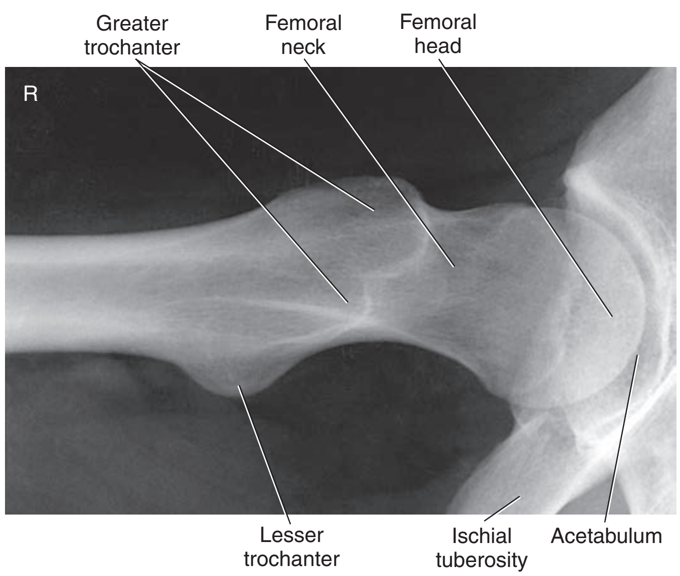


Fig. 7.77 Modified axiolateral projection.

7

RADIOGRAPHS FOR CRITIQUE

This section consists of an ideal projection (Image A) along with one or more projections that may demonstrate positioning and/or technical errors. Critique Figs. C7.78 through C7.81 Compare Image A to the other projections and identify the errors. While examining each image, consider the following questions:

1. Is all essential anatomy demonstrated on the image?
2. What positioning errors are present that compromise image quality?
3. Are technical factors optimal?
4. Is there evidence of collimation and are pre-exposure anatomic side markers visible on the image?
5. Do these errors require a repeat exposure?

Feedback for each set of images is located on the faculty Evolve site.

Fig. C7.78 AP pelvis projection.

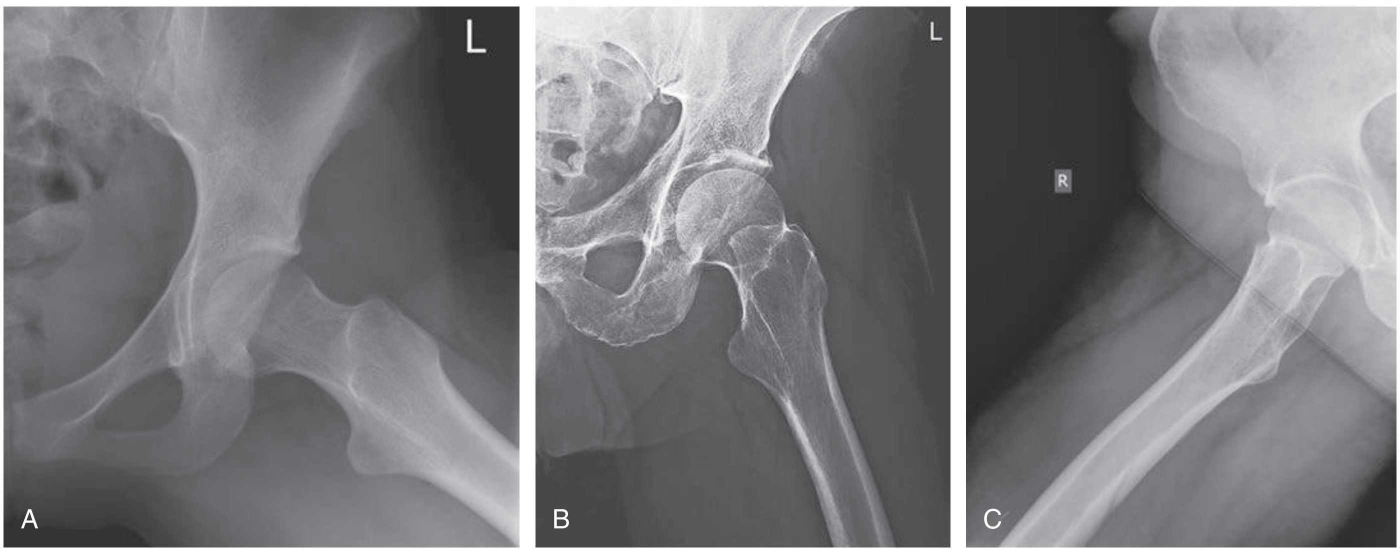


Fig. C7.79 Unilateral modified Cleaves method. (**A** from McQuillen Martensen K. *Radiographic image analysis*, ed 4, St. Louis, 2015, Saunders Elsevier; **C** from Dachs R et al. Double pathology, sarcoidosis associated with multiple myeloma: a case report. *Journal of Bone Oncology* 3(2):61–65.)

Fig. C7.80 Axiolateral (Danelius-Miller) projection. (**A** from Berry DJ. *Surgery of the hip,* ed 2, Philadelphia, 2020, Elsevier; **B** from Magee DJ. *Orthopedic physical assessment,* ed 6, Philadelphia, 2014, Saunders; **C** from Berry DJ. *Surgery of the hip,* Philadelphia, 2013, Saunders.)

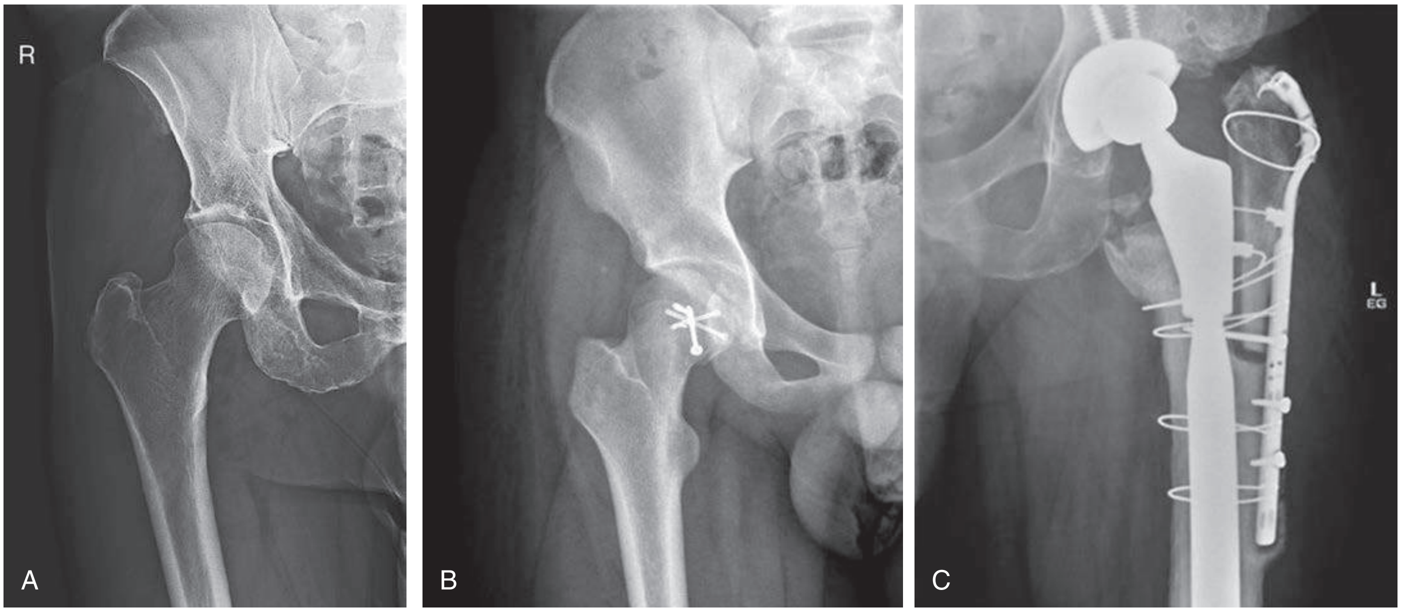


Fig. C7.81 **Unilateral AP hip projection.** (**B** from Ying LJ. A case of pathological fracture caused by vitamin D insufficiency in a young athlete and a review of the literature. *Journal of Clinical Orthopaedics and Trau*ma 10(6):1111–1115 **C** From Berry DJ. *Surgery of the hip,* ed 2, Philadelphia, 2020, Elsevier.)